Set 4E: BJT Amplifier Design

ECEN 325 — Lecture 4 (Bipolar Junction Transistors) practice set

What this set trains. Designing a complete amplifier from a specification table, with the design procedures of the lecture: pick V_{RE} for V_{BE}-insensitivity, the largest V_{RC} the output swing allows, R_H of the output follower from current clipping, then the largest bias current that still meets the input-resistance specification; every resistor follows from Ohm's law and the gain formula. Each solution ends with an exact analysis of the finished circuit ($\beta = 100$, base currents included) and, where needed, the small adjustment the slides do in simulation (trim R_{G1} for the exact gain, lower R_H if the follower would clip). Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-4 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1–3 tools):

T1 *Active BJT at DC:* npn $V_{BE} = 0.7\,\text{V}$, pnp $V_{EB} = 0.7\,\text{V}$; $I_C = \beta I_B$, $I_E = (\beta+1)I_B$, so $I_C \approx I_E$ and, when β is large, $I_B \approx 0$. Active needs $|V_{CE}| \geq 0.2\,\text{V}$.

T2 *Current mirror:* transistors with the same base node and the same emitter node have the same V_{BE}, hence the same I_C.

T3 *Small-signal parameters:* $r_e = \frac{V_T}{I_C}$ ($V_T = 25\,\text{mV}$), $\alpha = \frac{\beta}{\beta+1}$, $r_o = \infty$. For AC, DC supplies are ground and "∞" capacitors are shorts.

T4 *Node impedances and gains:* $Z_{base} = (\beta+1)(r_e + Z_E)$, $Z_{emitter} = r_e + \frac{Z_B}{\beta+1}$, $Z_{collector} = \infty$; CE $\frac{v_c}{v_b} = -\alpha\frac{Z_C}{r_e+Z_E}$, CB $\frac{v_c}{v_e} = \alpha\frac{Z_C}{Z_{emitter}}$, CC $\frac{v_e}{v_b} = \frac{Z_E}{r_e+Z_E}$. Z_C, Z_E, Z_B are everything else hanging on that node (including the input resistance of the next stage).

Steps:

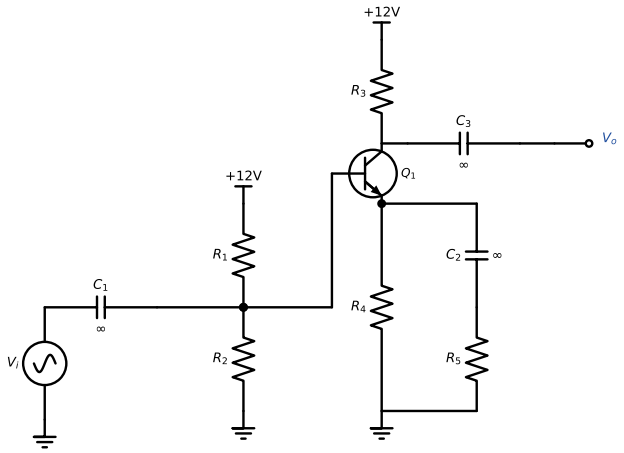
1. *DC:* assume every transistor active and $I_B \approx 0$; walk through the circuit node by node (dividers, V_{BE}, KCL for I_E, $I_C \approx I_E$, collector voltages); then *verify*: $I_B \ll$ bias currents and $|V_{CE}| \geq 0.2\,\text{V}$.
2. *AC:* r_e of every transistor; input resistance of every stage from the last one backwards; multiply the node-to-node gains along the signal path; then standard form, Bode plot and output as in Lecture 1.
3. *Design:* run the same relations backwards, choosing the free quantities with the stated design rules, then check the result by analysing the finished circuit.

Problems

Easy problems

Problem 4E.1 — Easy

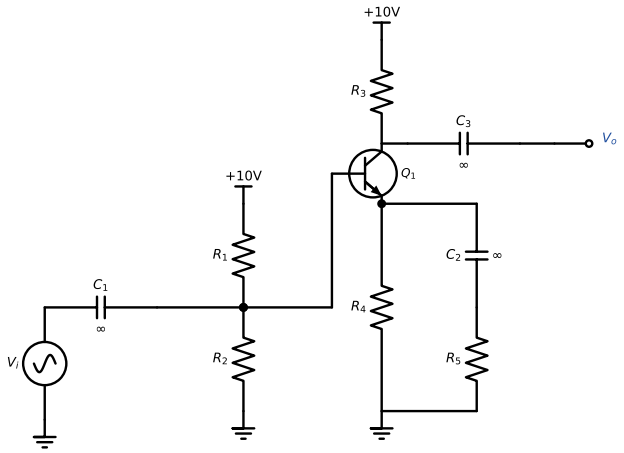
Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; $V_{RE1} = 1\,\text{V}$. Where a current is not given, take the nearest E12 value on the safe side.



specification	value
Supply voltage V_{CC}	$12\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 2\,\text{V}$
Voltage gain $\lvert A_v \rvert$	50
Input resistance R_i	$\geq 5\,\text{k}\Omega$
THD at $\hat{V}_o = 2\,\text{V}$	$\leq 2\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Problem 4E.2 — Easy

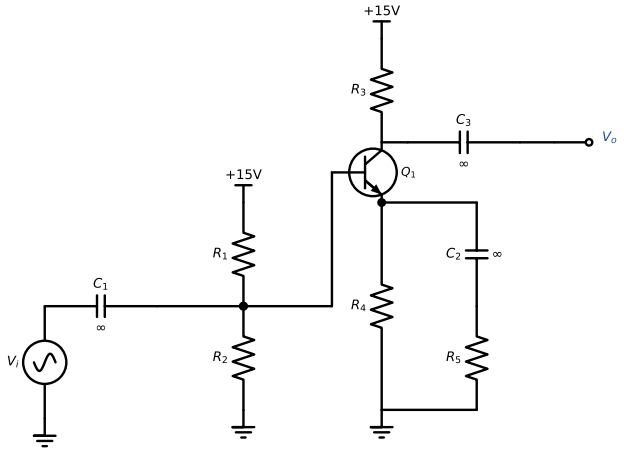
Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; $V_{RE1} = 1\,\text{V}$. Where a current is not given, take the nearest E12 value on the safe side.



specification	value
Supply voltage V_{CC}	$10\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 1.5\,\text{V}$
Voltage gain $\lvert A_v \rvert$	80
Input resistance R_i	$\geq 3\,\text{k}\Omega$
THD at $\hat{V}_o = 1.5\,\text{V}$	$\leq 3\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Problem 4E.3 — Easy

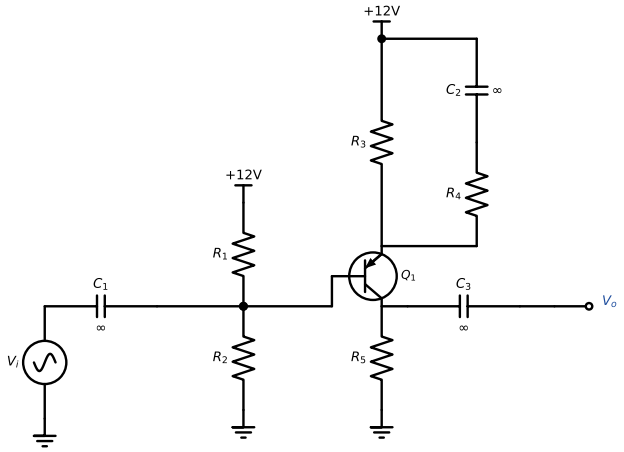
Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; $V_{RE1} = 1\,\text{V}$. Where a current is not given, take the nearest E12 value on the safe side.



specification	value
Supply voltage V_{CC}	$15\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 3\,\text{V}$
Voltage gain $\lvert A_v \rvert$	30
Input resistance R_i	$\geq 10\,\text{k}\Omega$
THD at $\hat{V}_o = 3\,\text{V}$	$\leq 2\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Problem 4E.4 — Easy

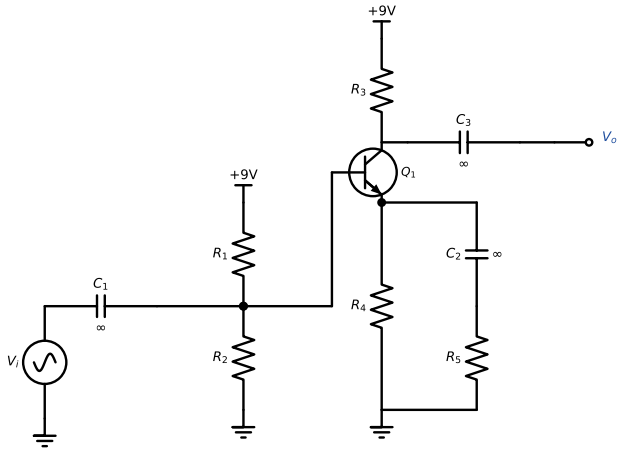
Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; $V_{RE1} = 1\,\text{V}$. Where a current is not given, take the nearest E12 value on the safe side.



specification	value
Supply voltage V_{CC}	$12\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 2\,\text{V}$
Voltage gain $\lvert A_v \rvert$	40
Input resistance R_i	$\geq 4\,\text{k}\Omega$
THD at $\hat{V}_o = 2\,\text{V}$	$\leq 2\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Problem 4E.5 — Easy

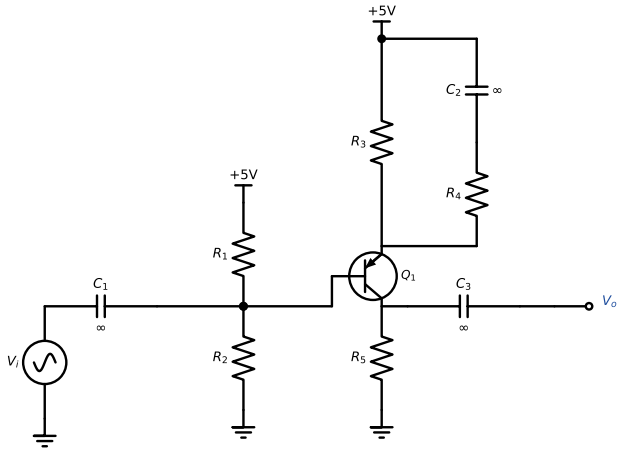
Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; $V_{RE1} = 1\,\text{V}$. Where a current is not given, take the nearest E12 value on the safe side.



specification	value
Supply voltage V_{CC}	$9\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 1\,\text{V}$
Voltage gain $\lvert A_v \rvert$	100
Input resistance R_i	$\geq 2\,\text{k}\Omega$
THD at $\hat{V}_o = 1\,\text{V}$	$\leq 3\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Problem 4E.6 — Easy

Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; $V_{RE1} = 1\,\text{V}$. Where a current is not given, take the nearest E12 value on the safe side.

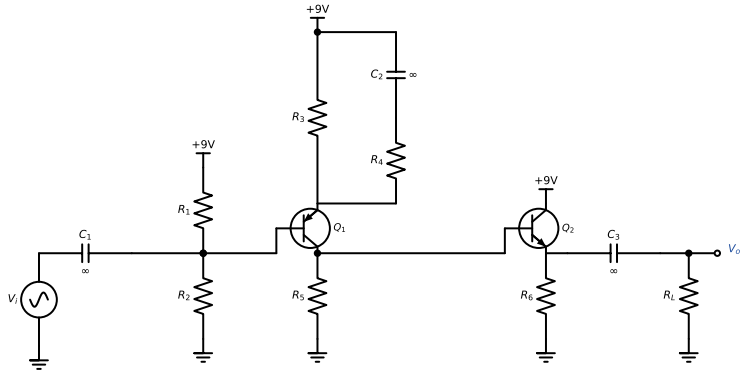


specification	value
Supply voltage V_{CC}	$5\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 1\,\text{V}$
Voltage gain $\lvert A_v \rvert$	50
Input resistance R_i	$\geq 1\,\text{k}\Omega$
THD at $\hat{V}_o = 1\,\text{V}$	$\leq 5\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Medium problems

Problem 4E.7 — Medium

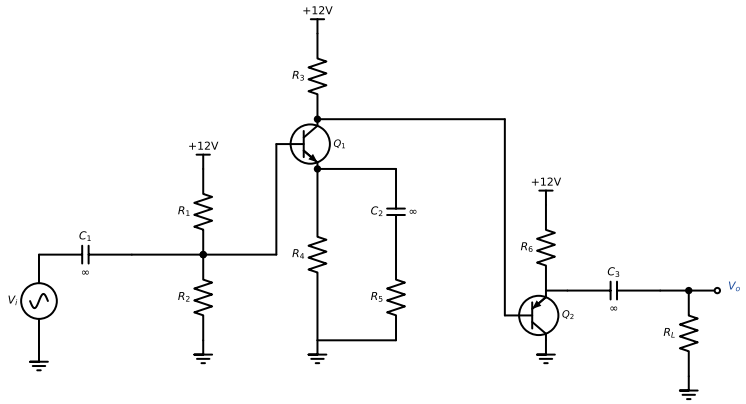
Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; design for $\hat{V}_o = 1.8\,\text{V}$ (margin); $V_{RE1} = 1\,\text{V}$. Where a current or R_H is not given, take the nearest E12 value on the safe side.



specification	value
Supply voltage V_{CC}	9 V
0-to-peak output swing $\hat{V}_o$	$\geq 1.5\,\text{V}$
Voltage gain $\lvert A_v \rvert$	40
Input resistance R_i	$\geq 4\,\text{k}\Omega$
Load resistance R_L	$500\,\Omega$
THD at $\hat{V}_o = 1.5\,\text{V}$	$\leq 3\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Problem 4E.8 — Medium

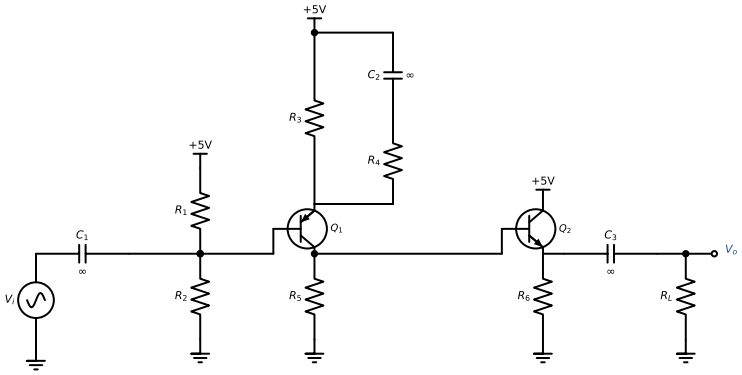
Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; design for $\hat{V}_o = 2.4\,\text{V}$ (margin); $V_{RE1} = 1\,\text{V}$. Where a current or R_H is not given, take the nearest E12 value on the safe side.



specification	value
Supply voltage V_{CC}	12 V
0-to-peak output swing $\hat{V}_o$	$\geq 2\,\text{V}$
Voltage gain $\lvert A_v \rvert$	60
Input resistance R_i	$\geq 5\,\text{k}\Omega$
Load resistance R_L	$1\,\text{k}\Omega$
THD at $\hat{V}_o = 2\,\text{V}$	$\leq 3\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Problem 4E.9 — Medium

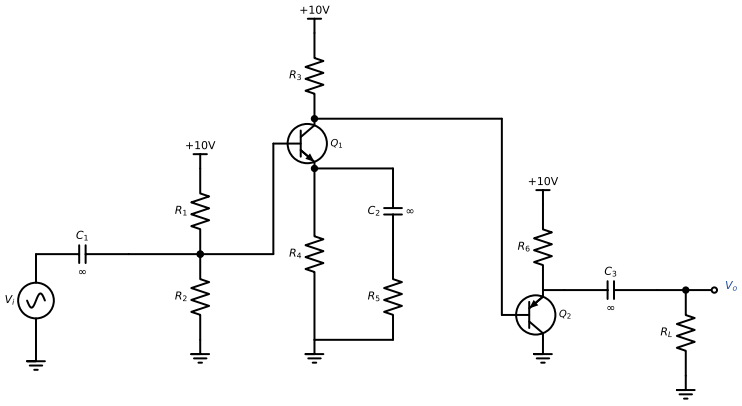
Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; design for $\hat{V}_o = 1.2\,\text{V}$ (margin); $V_{RE1} = 1\,\text{V}$. Where a current or R_H is not given, take the nearest E12 value on the safe side.



specification	value
Supply voltage V_{CC}	$5\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 1\,\text{V}$
Voltage gain $\lvert A_v \rvert$	30
Input resistance R_i	$\geq 1.5\,\text{k}\Omega$
Load resistance R_L	$200\,\Omega$
THD at $\hat{V}_o = 1\,\text{V}$	$\leq 5\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Problem 4E.10 — Medium

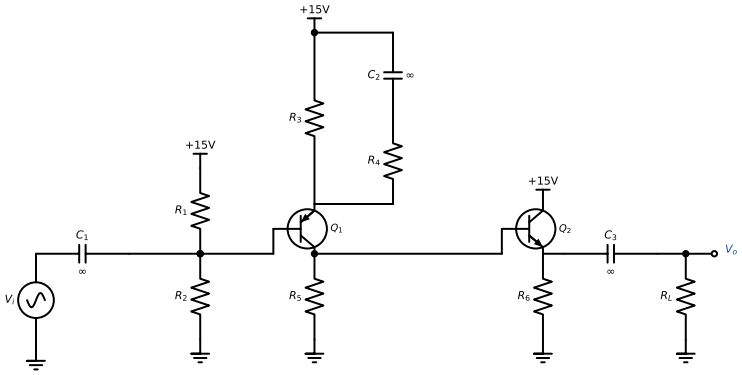
Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; design for $\hat{V}_o = 1.8\,\text{V}$ (margin); $V_{RE1} = 1\,\text{V}$. Where a current or R_H is not given, take the nearest E12 value on the safe side.



specification	value
Supply voltage V_{CC}	$10\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 1.5\,\text{V}$
Voltage gain $\lvert A_v \rvert$	100
Input resistance R_i	$\geq 3\,\text{k}\Omega$
Load resistance R_L	$300\,\Omega$
THD at $\hat{V}_o = 1.5\,\text{V}$	$\leq 5\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Problem 4E.11 — Medium

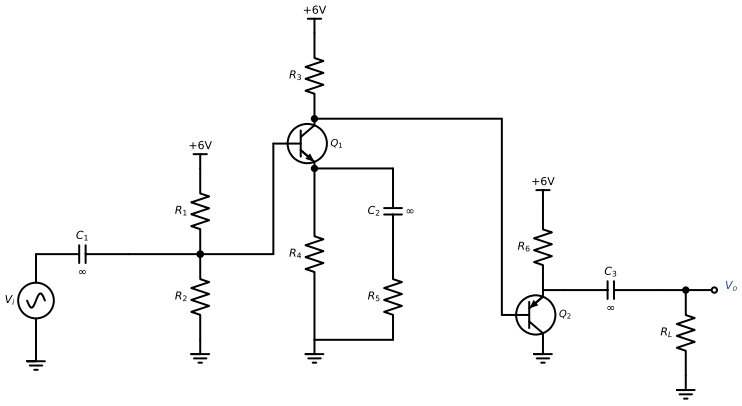
Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 20$; design for $\hat{V}_o = 3.5\,\text{V}$ (margin); $V_{RE1} = 1\,\text{V}$. Where a current or R_H is not given, take the nearest E12 value on the safe side.



specification	value
Supply voltage V_{CC}	$15\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 3\,\text{V}$
Voltage gain $\lvert A_v \rvert$	30
Input resistance R_i	$\geq 10\,\text{k}\Omega$
Load resistance R_L	$1\,\text{k}\Omega$
THD at $\hat{V}_o = 3\,\text{V}$	$\leq 3\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Problem 4E.12 — Medium

Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; design for $\hat{V}_o = 1.2\,\text{V}$ (margin); $V_{RE1} = 1\,\text{V}$. Where a current or R_H is not given, take the nearest E12 value on the safe side.



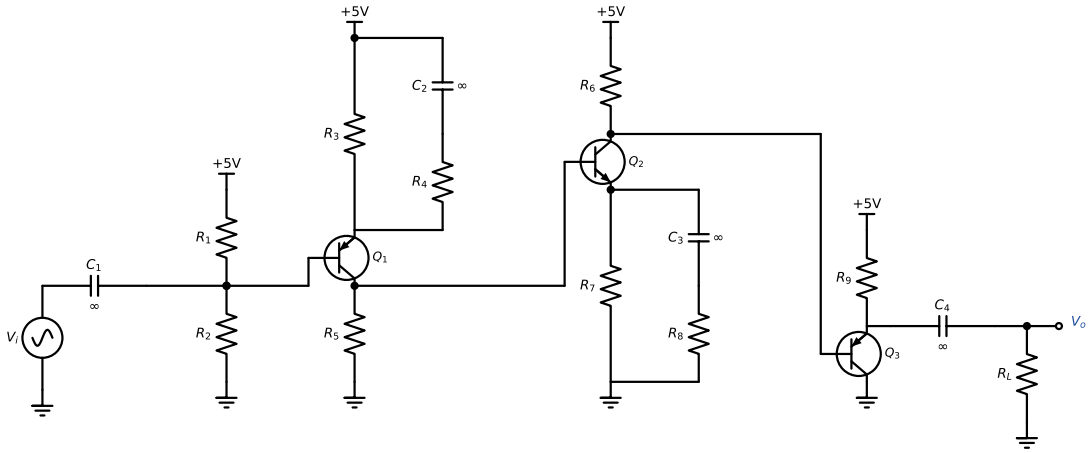
specification	value
Supply voltage V_{CC}	$6\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 1\,\text{V}$
Voltage gain $\lvert A_v \rvert$	25
Input resistance R_i	$\geq 2\,\text{k}\Omega$
Load resistance R_L	$100\,\Omega$
THD at $\hat{V}_o = 1\,\text{V}$	$\leq 3\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$

Hard problems

Problem 4E.13 **Hard**

Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 20$; design for $\hat{V}_o = 1.3\,\text{V}$ (margin); $V_{RE1} = 1.8\,\text{V}$, $V_{RE2} = 1\,\text{V}$; $|A_{v2}| = 6$; $I_{C2} = 8\,\text{mA}$; $R_H = 50\,\Omega$. Where a current or R_H is not given, take the nearest E12 value on the safe side. (These are the specifications of Sample Problem 11.)

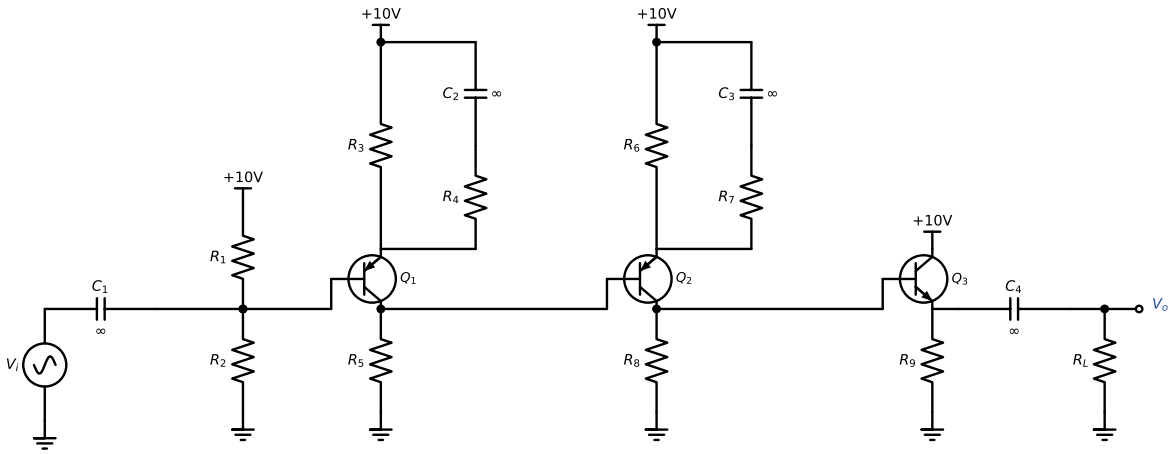
specification	value
Supply voltage V_{CC}	$5\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 1\,\text{V}$
Voltage gain $\lvert A_v \rvert$	100
Input resistance R_i	$\geq 2\,\text{k}\Omega$
Load resistance R_L	$150\,\Omega$
THD at $\hat{V}_o = 1\,\text{V}$	$\leq 3\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$



Problem 4E.14 **Hard**

Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; design for $\hat{V}_o = 2.4\,\text{V}$ (margin); $V_{RE1} = 1\,\text{V}$, $V_{RE2} = 1\,\text{V}$; $|A_{v2}| = 10$. Where a current or R_H is not given, take the nearest E12 value on the safe side.

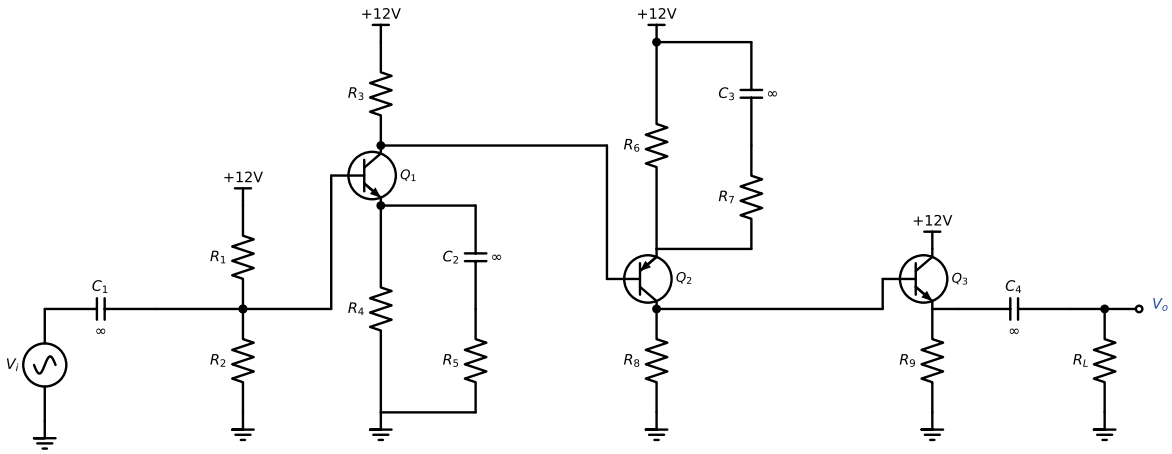
specification	value
Supply voltage V_{CC}	$10\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 2\,\text{V}$
Voltage gain $\lvert A_v \rvert$	200
Input resistance R_i	$\geq 5\,\text{k}\Omega$
Load resistance R_L	$500\,\Omega$
THD at $\hat{V}_o = 2\,\text{V}$	$\leq 5\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$



Problem 4E.15 **Hard**

Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 20$; design for $\hat{V}_o = 2.4\,\text{V}$ (margin); $V_{RE1} = 1.5\,\text{V}$, $V_{RE2} = 1\,\text{V}$; $|A_{v2}| = 8$. Where a current or R_H is not given, take the nearest E12 value on the safe side.

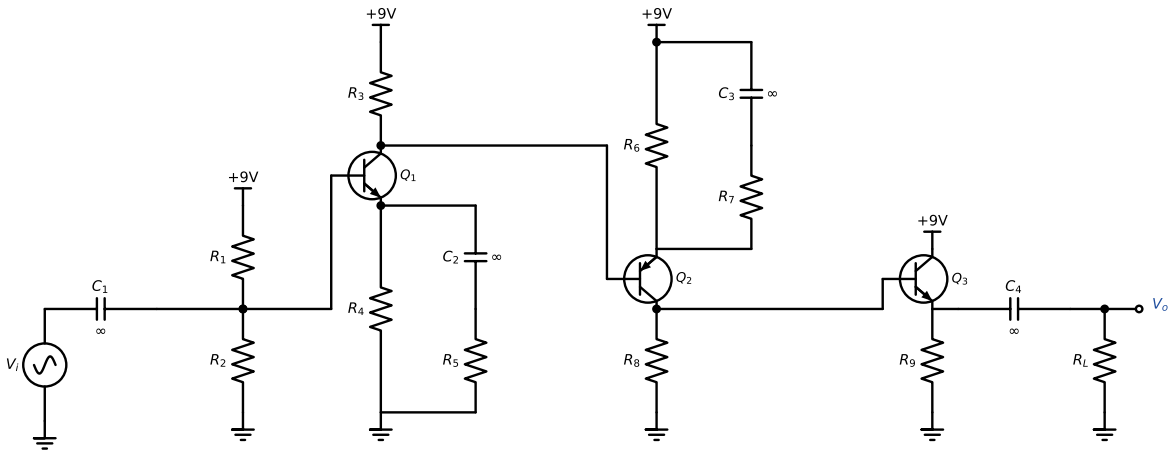
specification	value
Supply voltage V_{CC}	$12\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 2\,\text{V}$
Voltage gain $\lvert A_v \rvert$	150
Input resistance R_i	$\geq 4\,\text{k}\Omega$
Load resistance R_L	$1\,\text{k}\Omega$
THD at $\hat{V}_o = 2\,\text{V}$	$\leq 5\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$



Problem 4E.16 **Hard**

Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; design for $\hat{V}_o = 1.8\,\text{V}$ (margin); $V_{RE1} = 1\,\text{V}$, $V_{RE2} = 1\,\text{V}$; $|A_{v2}| = 8$. Where a current or R_H is not given, take the nearest E12 value on the safe side.

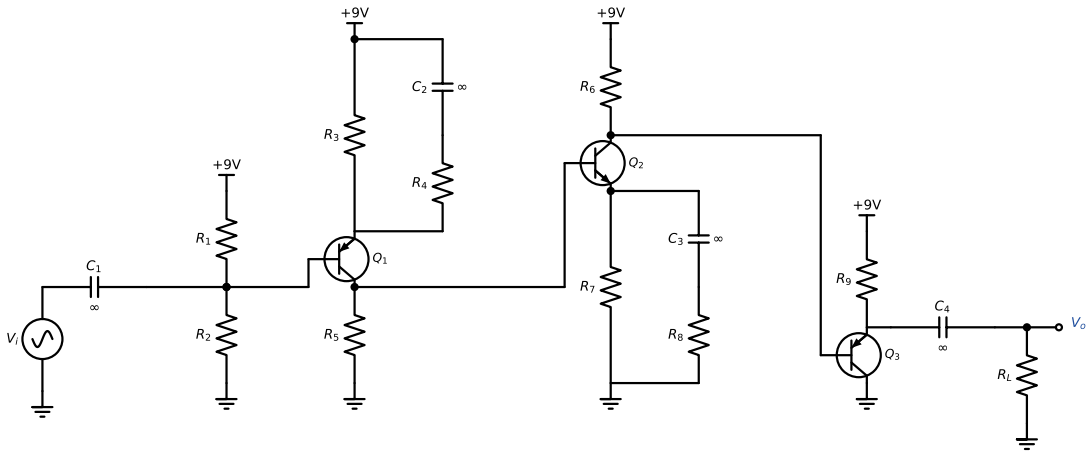
specification	value
Supply voltage V_{CC}	$9\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 1.5\,\text{V}$
Voltage gain $\lvert A_v \rvert$	120
Input resistance R_i	$\geq 3\,\text{k}\Omega$
Load resistance R_L	$300\,\Omega$
THD at $\hat{V}_o = 1.5\,\text{V}$	$\leq 5\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$



Problem 4E.17 **Hard**

Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 20$; design for $\hat{V}_o = 2.5\,\text{V}$ (margin); $V_{RE1} = 1.5\,\text{V}$, $V_{RE2} = 1\,\text{V}$; $|A_{v2}| = 8$. Where a current or R_H is not given, take the nearest E12 value on the safe side.

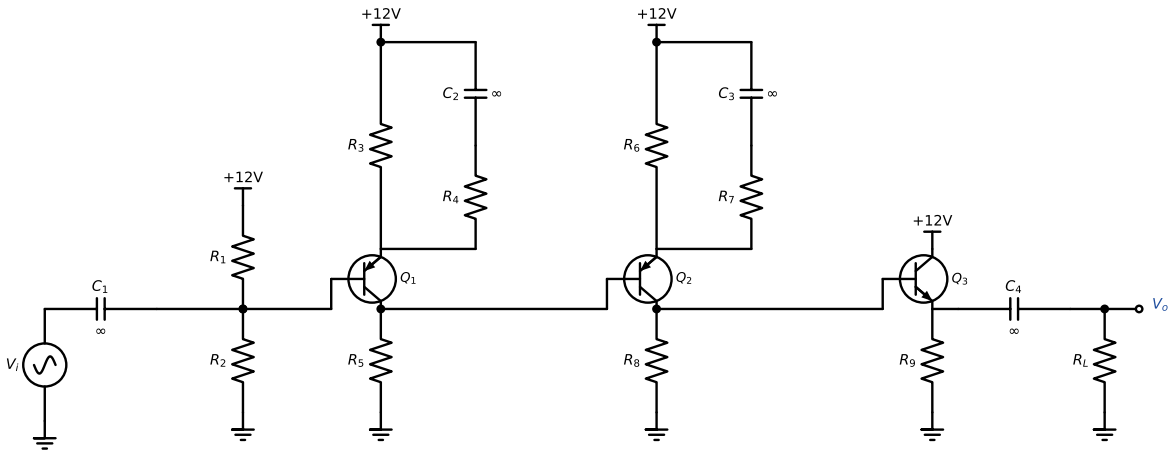
specification	value
Supply voltage V_{CC}	$9\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 2\,\text{V}$
Voltage gain $\lvert A_v \rvert$	150
Input resistance R_i	$\geq 2\,\text{k}\Omega$
Load resistance R_L	$200\,\Omega$
THD at $\hat{V}_o = 2\,\text{V}$	$\leq 5\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$



Problem 4E.18 **Hard**

Design the amplifier below (find all resistor values) for the specifications in the table, insensitive to β and V_{BE} variations. $\beta = 100$, $V_T = 25\,\text{mV}$, $|V_{BE}| = 0.7\,\text{V}$, $V_{CE,sat} = 0.2\,\text{V}$; capacitors marked ∞ are shorts at the signal frequency. Use: $N = 10$; design for $\hat{V}_o = 3.5\,\text{V}$ (margin); $V_{RE1} = 1\,\text{V}$, $V_{RE2} = 1\,\text{V}$; $|A_{v2}| = 12$. Where a current or R_H is not given, take the nearest E12 value on the safe side.

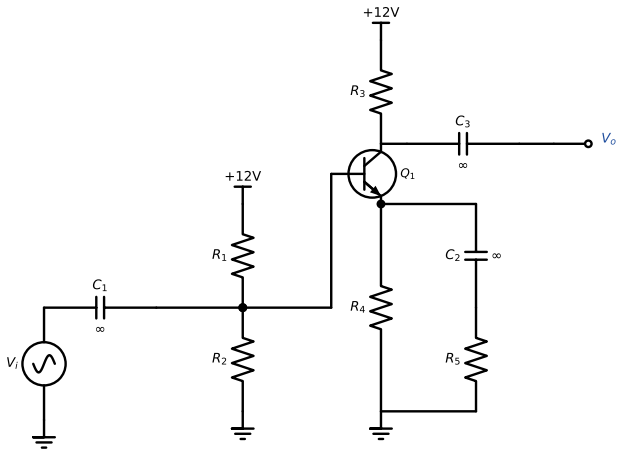
specification	value
Supply voltage V_{CC}	$12\,\text{V}$
0-to-peak output swing $\hat{V}_o$	$\geq 3\,\text{V}$
Voltage gain $\lvert A_v \rvert$	300
Input resistance R_i	$\geq 2\,\text{k}\Omega$
Load resistance R_L	$1\,\text{k}\Omega$
THD at $\hat{V}_o = 3\,\text{V}$	$\leq 5\,\%$
$\Delta I_C / I_C$ for $\lvert V_{BE} \rvert = 0.7 \pm 0.1\,\text{V}$	$\leq 10\,\%$



Solutions

Every solution follows the steps of the Lecture-4 pipeline card.

Solution 4E.1 — Easy



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{C1}$, $R_4 = R_{E1}$, $R_5 = R_{G1}$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 12 - 2 - 1 - 0.2 = 8.8\,\text{V} \;\geq\; \hat{V}_o = 2\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{8.8}{0.025} = 352 \geq 50$ ✓

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 1.7\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 10.3\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta/R_{i,spec}}{\frac{10}{10.3} + \frac{10}{1.7} + \frac{50}{8.8}} = 1.6\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1.5\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{8.8}{1.5\,\text{mA}} = \mathbf{5.87\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{1.5\,\text{mA}} = \mathbf{667\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 16.7\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}}{r_{e1}+R_{E1}\|R_{G1}} = 50$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}}{|A_{v1}|} - r_{e1} = 101\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{101\,\Omega} - \frac{1}{667\,\Omega}} = \mathbf{119\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{10.3}{150\,\mu\text{A}} = \mathbf{68.7\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{1.7}{150\,\mu\text{A}} = \mathbf{11.3\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1}+R_{E1}\|R_{G1}) = 68.7\,\text{k}\Omega\|11.3\,\text{k}\Omega\|11.9\,\text{k}\Omega = 5.342\,\text{k}\Omega \geq 5\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\|V_{CE}\|$
Q_1	1.5 mA	1.3 mA	3.513 V

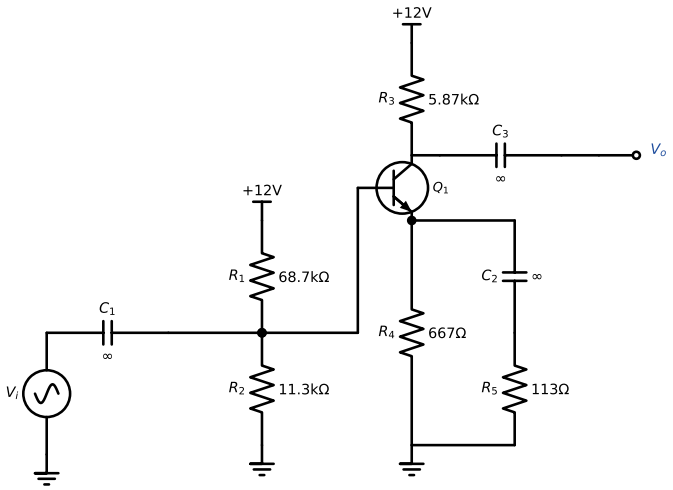
The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 48.43$ instead of 50 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1}+R_{E1}\|R_{G1})$, so scale that sum by 48.43/50 (exact $r_{e1} = 19.3\,\Omega$):

$$R_{E1}\|R_{G1} = (19.3\,\Omega + 101\,\Omega)\frac{48.43}{50} - 19.3\,\Omega = 96.9\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{113\,\Omega}$$

Re-analysis: $|A_v| = 50$ ✓.

- Input resistance (exact): $R_i = 5.32\,\text{k}\Omega \geq 5\,\text{k}\Omega$ ✓
- Output swing: Q_1 clips (saturation or cutoff) at $\hat{V}_o = 3.313\,\text{V}$. The smallest, 3.313 V, is $\geq 2\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 1\,\% \leq 2\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

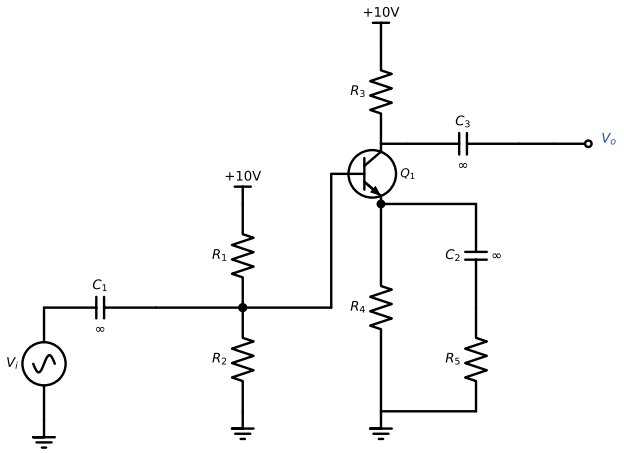
resistor	value
R_1 (R_{B1})	68.7 kΩ
R_2 (R_{B2})	11.3 kΩ
R_3 (R_{C1})	5.87 kΩ
R_4 (R_{E1})	667 Ω
R_5 (R_{G1})	113 Ω



Answer

$R_1 = 68.7\,\text{k}\Omega, \quad R_2 = 11.3\,\text{k}\Omega, \quad R_3 = 5.87\,\text{k}\Omega, \quad R_4 = 667\,\Omega, \quad R_5 = 113\,\Omega$

Solution 4E.2 **Easy**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{C1}$, $R_4 = R_{E1}$, $R_5 = R_{G1}$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 10 - 1.5 - 1 - 0.2 = 7.3\,\text{V} \;\geq\; \hat{V}_o = 1.5\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{7.3}{0.025} = 292 \geq 80$ ✓

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 1.7\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 8.3\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta/R_{i,spec}}{\frac{10}{8.3} + \frac{10}{1.7} + \frac{80}{7.3}} = 1.85\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1.8\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{7.3}{1.8\,\text{mA}} = \mathbf{4.06\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{1.8\,\text{mA}} = \mathbf{556\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 13.9\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}}{r_{e1} + R_{E1}\|R_{G1}} = 80$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}}{|A_{v1}|} - r_{e1} = 36.8\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{36.8\,\Omega} - \frac{1}{556\,\Omega}} = \mathbf{39.4\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{8.3}{180\,\mu\text{A}} = \mathbf{46.1\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{1.7}{180\,\mu\text{A}} = \mathbf{9.44\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 46.1\,\text{k}\Omega\|9.44\,\text{k}\Omega\|5.12\,\text{k}\Omega = 3.097\,\text{k}\Omega \geq 3\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	1.8 mA	1.56 mA	2.781 V

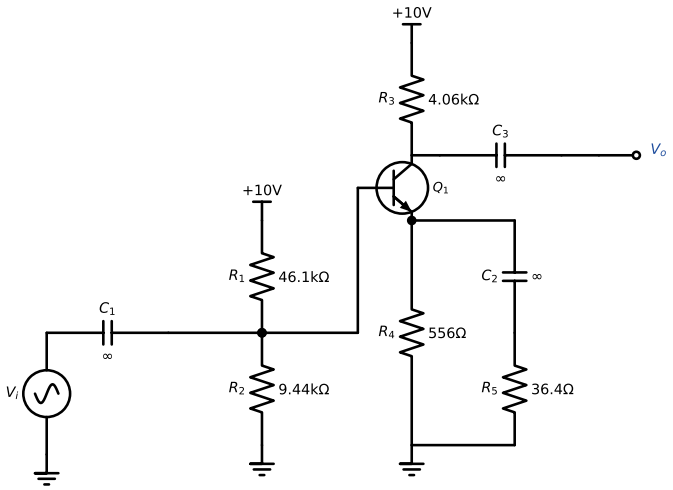
The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 76.06$ instead of 80 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 76.06/80 (exact $r_{e1} = 16\,\Omega$):

$$R_{E1}\|R_{G1} = (16\,\Omega + 36.8\,\Omega)\frac{76.06}{80} - 16\,\Omega = 34.2\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{36.4\,\Omega}$$

Re-analysis: $|A_v| = 80$ ✓.

- Input resistance (exact): $R_i = 3.08\,\text{k}\Omega \geq 3\,\text{k}\Omega$ ✓
- Output swing: Q_1 clips (saturation or cutoff) at $\hat{V}_o = 2.581\,\text{V}$. The smallest, 2.581 V, is $\geq 1.5\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 1.7\,\% \leq 3\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

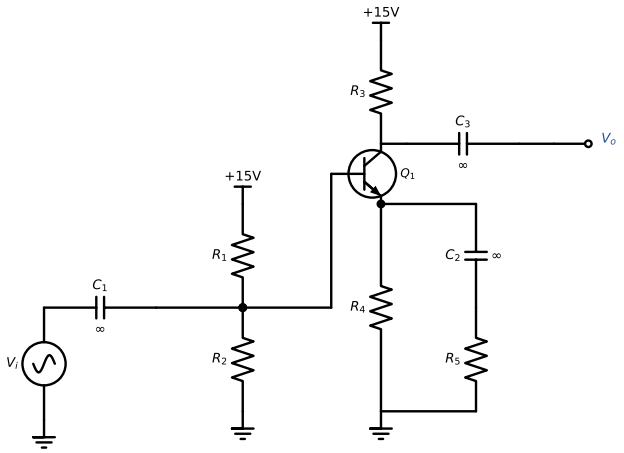
resistor	value
R_1 (R_{B1})	46.1 kΩ
R_2 (R_{B2})	9.44 kΩ
R_3 (R_{C1})	4.06 kΩ
R_4 (R_{E1})	556 Ω
R_5 (R_{G1})	36.4 Ω



Answer

$R_1 = 46.1\,\mathrm{k\Omega}, \quad R_2 = 9.44\,\mathrm{k\Omega}, \quad R_3 = 4.06\,\mathrm{k\Omega}, \quad R_4 = 556\,\Omega, \quad R_5 = 36.4\,\Omega$

Solution 4E.3 **Easy**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{C1}$, $R_4 = R_{E1}$, $R_5 = R_{G1}$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 15 - 3 - 1 - 0.2 = 10.8\,\text{V} \;\geq\; \hat{V}_o = 3\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{10.8}{0.025} = 432 \geq 30$ ✓

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 1.7\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 13.3\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta/R_{i,spec}}{\frac{10}{13.3} + \frac{10}{1.7} + \frac{30}{10.8}} = 1.06\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{10.8}{1\,\text{mA}} = \mathbf{10.8\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{1\,\text{mA}} = \mathbf{1\,k\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 25\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}}{r_{e1} + R_{E1}\|R_{G1}} = 30$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}}{|A_{v1}|} - r_{e1} = 335\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{335\,\Omega} - \frac{1}{1\,\text{k}\Omega}} = \mathbf{504\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{13.3}{100\,\mu\text{A}} = \mathbf{133\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{1.7}{100\,\mu\text{A}} = \mathbf{17\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 133\,\text{k}\Omega\|17\,\text{k}\Omega\|36.4\,\text{k}\Omega = 10.66\,\text{k}\Omega \geq 10\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	1 mA	862 μA	4.825 V

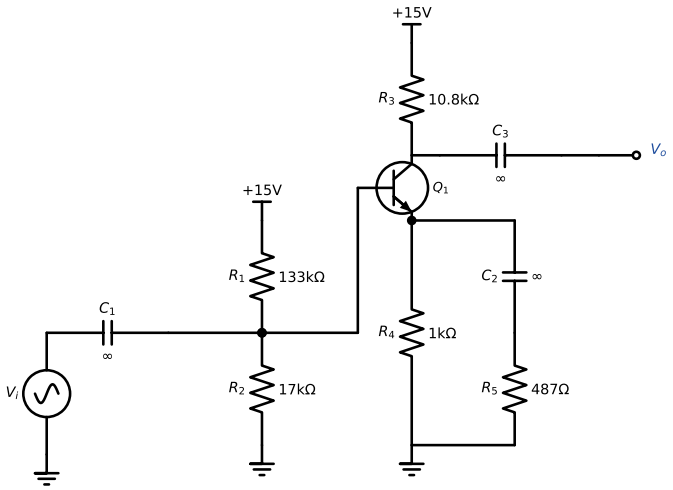
The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 29.38$ instead of 30 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 29.38/30 (exact $r_{e1} = 29\,\Omega$):

$$R_{E1}\|R_{G1} = (29\,\Omega + 335\,\Omega)\frac{29.38}{30} - 29\,\Omega = 327\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{487\,\Omega}$$

Re-analysis: $|A_v| = 30$ ✓.

- Input resistance (exact): $R_i = 10.6\,\text{k}\Omega \geq 10\,\text{k}\Omega$ ✓
- Output swing: Q_1 clips (saturation or cutoff) at $\hat{V}_o = 4.625\,\text{V}$. The smallest, 4.625 V, is $\geq 3\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 0.64\,\% \leq 2\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

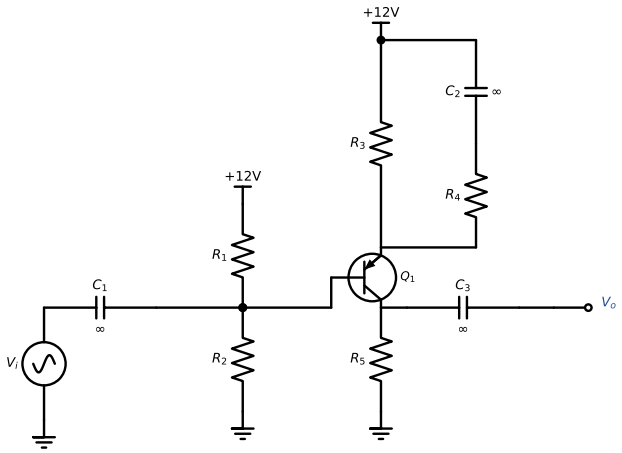
resistor	value
R_1 (R_{B1})	133 kΩ
R_2 (R_{B2})	17 kΩ
R_3 (R_{C1})	10.8 kΩ
R_4 (R_{E1})	1 kΩ
R_5 (R_{G1})	487 Ω



Answer

$R_1 = 133\,\text{k}\Omega, \quad R_2 = 17\,\text{k}\Omega, \quad R_3 = 10.8\,\text{k}\Omega, \quad R_4 = 1\,\text{k}\Omega, \quad R_5 = 487\,\Omega$

Solution 4E.4 **Easy**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{E1}$, $R_4 = R_{G1}$, $R_5 = R_{C1}$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 12 - 2 - 1 - 0.2 = 8.8\,\text{V} \;\geq\; \hat{V}_o = 2\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{8.8}{0.025} = 352 \geq 40$ ✓

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 10.3\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 1.7\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta/R_{i,spec}}{\frac{10}{1.7} + \frac{10}{10.3} + \frac{40}{8.8}} = 2.19\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1.8\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{8.8}{1.8\,\text{mA}} = \mathbf{4.89\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{1.8\,\text{mA}} = \mathbf{556\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 13.9\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}}{r_{e1} + R_{E1}\|R_{G1}} = 40$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}}{|A_{v1}|} - r_{e1} = 108\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{108\,\Omega} - \frac{1}{556\,\Omega}} = \mathbf{135\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{1.7}{180\,\mu\text{A}} = \mathbf{9.44\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{10.3}{180\,\mu\text{A}} = \mathbf{57.2\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 9.44\,\text{k}\Omega\|57.2\,\text{k}\Omega\|12.3\,\text{k}\Omega = 4.893\,\text{k}\Omega \geq 4\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	1.8 mA	1.56 mA	3.513 V

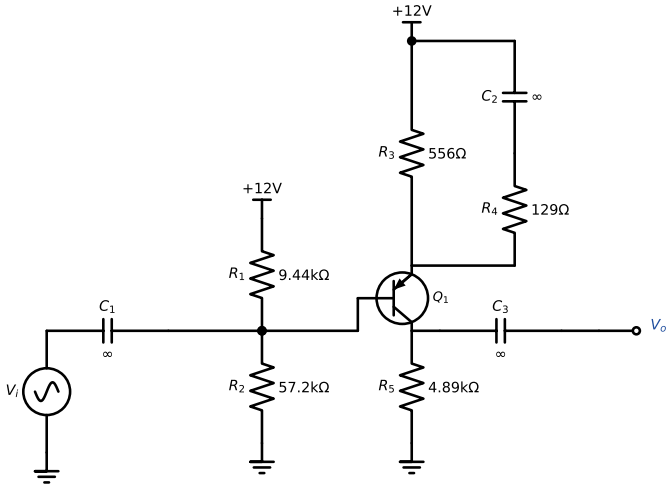
The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 38.91$ instead of 40 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 38.91/40 (exact $r_{e1} = 16.1\,\Omega$):

$$R_{E1}\|R_{G1} = (16.1\,\Omega + 108\,\Omega)\frac{38.91}{40} - 16.1\,\Omega = 105\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{129\,\Omega}$$

Re-analysis: $|A_v| = 40$ ✓.

- Input resistance (exact): $R_i = 4.87\,\text{k}\Omega \geq 4\,\text{k}\Omega$ ✓
- Output swing: Q_1 clips (saturation or cutoff) at $\hat{V}_o = 3.313\,\text{V}$. The smallest, 3.313 V, is $\geq 2\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e + R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T |A_v|}{4V_{RC}^2}$): THD $\approx 0.83\,\% \leq 2\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

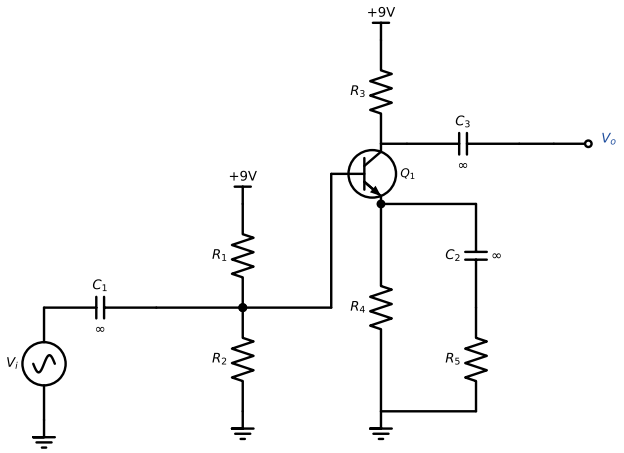
resistor	value
R_1 (R_{B1})	9.44 kΩ
R_2 (R_{B2})	57.2 kΩ
R_3 (R_{E1})	556 Ω
R_4 (R_{G1})	129 Ω
R_5 (R_{C1})	4.89 kΩ



Answer

$R_1 = 9.44\,\mathrm{k\Omega}, \quad R_2 = 57.2\,\mathrm{k\Omega}, \quad R_3 = 556\,\Omega, \quad R_4 = 129\,\Omega, \quad R_5 = 4.89\,\mathrm{k\Omega}$

Solution 4E.5 **Easy**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{C1}$, $R_4 = R_{E1}$, $R_5 = R_{G1}$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 9 - 1 - 1 - 0.2 = 6.8\,\text{V} \;\geq\; \hat{V}_o = 1\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{6.8}{0.025} = 272 \geq 100\;\checkmark$

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 1.7\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 7.3\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta/R_{i,spec}}{\frac{10}{7.3} + \frac{10}{1.7} + \frac{100}{6.8}} = 2.28\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 2.2\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{6.8}{2.2\,\text{mA}} = \mathbf{3.09\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{2.2\,\text{mA}} = \mathbf{455\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 11.4\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}}{r_{e1} + R_{E1}\|R_{G1}} = 100$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}}{|A_{v1}|} - r_{e1} = 19.5\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{19.5\,\Omega} - \frac{1}{455\,\Omega}} = \mathbf{20.4\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{7.3}{220\,\mu\text{A}} = \mathbf{33.2\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{1.7}{220\,\mu\text{A}} = \mathbf{7.73\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 33.2\,\text{k}\Omega\|7.73\,\text{k}\Omega\|3.12\,\text{k}\Omega = 2.084\,\text{k}\Omega \geq 2\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	2.2 mA	1.92 mA	2.196 V

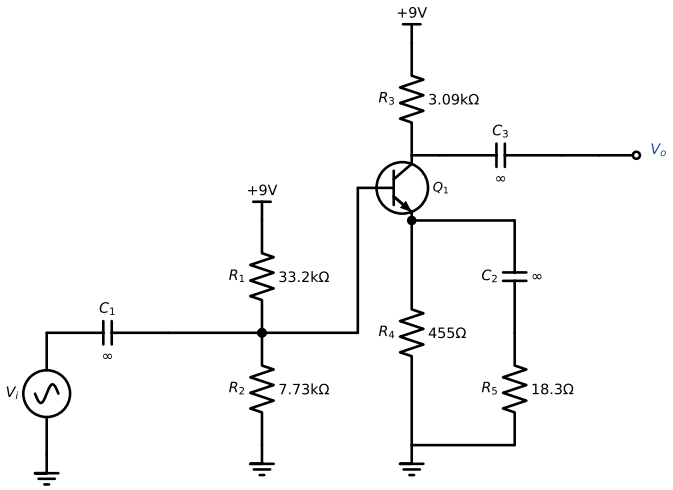
The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 93.9$ instead of 100 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 93.9/100 (exact $r_{e1} = 13\,\Omega$):

$$R_{E1}\|R_{G1} = (13\,\Omega + 19.5\,\Omega)\frac{93.9}{100} - 13\,\Omega = 17.6\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{18.3\,\Omega}$$

Re-analysis: $|A_v| = 100$ ✓.

- Input resistance (exact): $R_i = 2.07\,\text{k}\Omega \geq 2\,\text{k}\Omega$ ✓
- Output swing: Q_1 clips (saturation or cutoff) at $\hat{V}_o = 1.996\,\text{V}$. The smallest, 1.996 V, is $\geq 1\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 1.6\,\% \leq 3\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

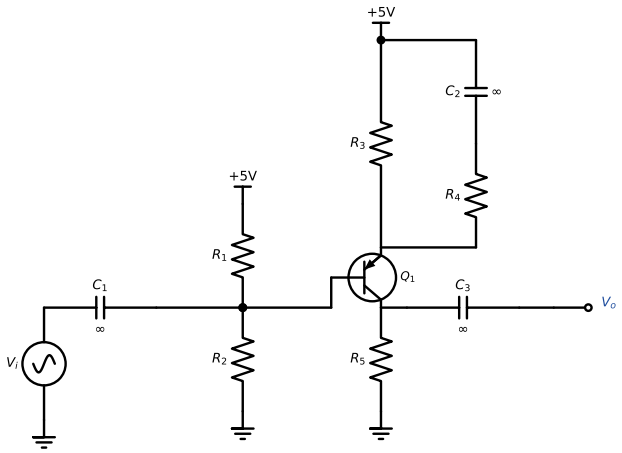
resistor	value
R_1 (R_{B1})	33.2 kΩ
R_2 (R_{B2})	7.73 kΩ
R_3 (R_{C1})	3.09 kΩ
R_4 (R_{E1})	455 Ω
R_5 (R_{G1})	18.3 Ω



Answer

$R_1 = 33.2\,\text{k}\Omega, \quad R_2 = 7.73\,\text{k}\Omega, \quad R_3 = 3.09\,\text{k}\Omega, \quad R_4 = 455\,\Omega, \quad R_5 = 18.3\,\Omega$

Solution 4E.6 **Easy**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{E1}$, $R_4 = R_{G1}$, $R_5 = R_{C1}$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 5 - 1 - 1 - 0.2 = 2.8\,\text{V} \;\geq\; \hat{V}_o = 1\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{2.8}{0.025} = 112 \geq 50$ ✓

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 3.3\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 1.7\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta/R_{i,spec}}{\frac{10}{1.7} + \frac{10}{3.3} + \frac{50}{2.8}} = 3.74\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 3.3\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{2.8}{3.3\,\text{mA}} = \mathbf{848\,\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{3.3\,\text{mA}} = \mathbf{303\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 7.58\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}}{r_{e1} + R_{E1}\|R_{G1}} = 50$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}}{|A_{v1}|} - r_{e1} = 9.39\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{9.39\,\Omega} - \frac{1}{303\,\Omega}} = \mathbf{9.69\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{1.7}{330\,\mu\text{A}} = \mathbf{5.15\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{3.3}{330\,\mu\text{A}} = \mathbf{10\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 5.15\,\text{k}\Omega\|10\,\text{k}\Omega\|1.71\,\text{k}\Omega = 1.14\,\text{k}\Omega \geq 1\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	3.3 mA	2.94 mA	1.605 V

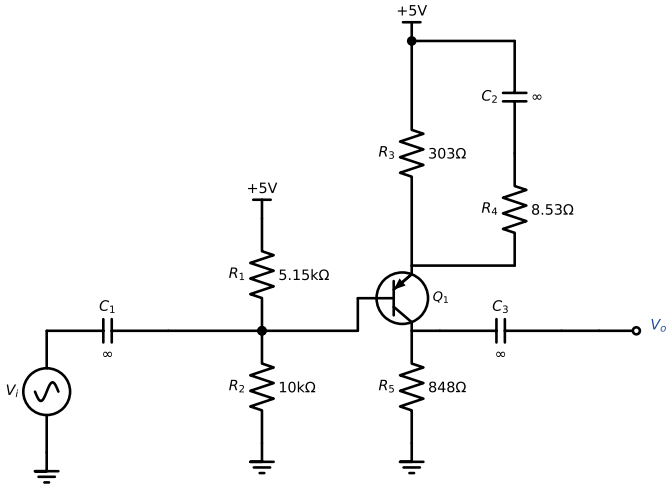
The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 46.94$ instead of 50 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 46.94/50 (exact $r_{e1} = 8.5\,\Omega$):

$$R_{E1}\|R_{G1} = (8.5\,\Omega + 9.39\,\Omega)\frac{46.94}{50} - 8.5\,\Omega = 8.3\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{8.53\,\Omega}$$

Re-analysis: $|A_v| = 50$ ✓.

- Input resistance (exact): $R_i = 1.13\,\text{k}\Omega \geq 1\,\text{k}\Omega$ ✓
- Output swing: Q_1 clips (saturation or cutoff) at $\hat{V}_o = 1.405\,\text{V}$. The smallest, 1.405 V, is $\geq 1\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e + R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T |A_v|}{4V_{RC}^2}$): THD $\approx 4.5\,\% \leq 5\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

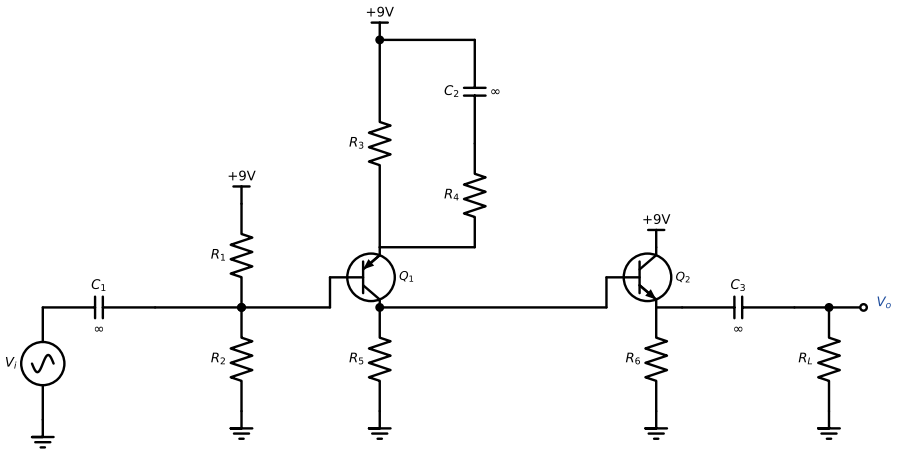
resistor	value
R_1 (R_{B1})	5.15 kΩ
R_2 (R_{B2})	10 kΩ
R_3 (R_{E1})	303 Ω
R_4 (R_{G1})	8.53 Ω
R_5 (R_{C1})	848 Ω



Answer

$R_1 = 5.15\,\text{k}\Omega, \quad R_2 = 10\,\text{k}\Omega, \quad R_3 = 303\,\Omega, \quad R_4 = 8.53\,\Omega, \quad R_5 = 848\,\Omega$

Solution 4E.7 **Medium**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{E1}$, $R_4 = R_{G1}$, $R_5 = R_{C1}$, $R_6 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 9 - 1.8 - 1 - 0.2 = 6\,\text{V} \;\geq\; \hat{V}_o = 1.8\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{6}{0.025} = 240 \geq 40$ ✓

Follower Q_2: its emitter sits at 0.7 V from the collector of Q_1, so $V_{RH} = V_{RC1} - 0.7 = 5.3\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC1} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC1} - 0.7 - \hat{V}_o}{\hat{V}_o} = 500\,\Omega \cdot \frac{5.3 - 1.8}{1.8} = 972\,\Omega$$

Take the E12 value just below: $R_H = \mathbf{820\,\Omega}$. Then $I_{C2} \approx \frac{V_{RH}}{R_H} = \frac{5.3}{820\,\Omega} = 6.46\,\text{mA}$, $r_{e2} = \frac{V_T}{I_{C2}} = 3.87\,\Omega$, and with $R_H \| R_L = 311\,\Omega$:

$$A_{v2} = \frac{R_H \| R_L}{r_{e2} + R_H \| R_L} = 0.9877, \qquad R_{i2} = (\beta+1)(r_{e2} + R_H \| R_L) = 31.8\,\text{k}\Omega$$

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 7.3\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 1.7\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E \| R_G) \approx \beta \frac{R_{C1}}{|A_{v1}|} = \beta \frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta/R_{i,spec}}{\frac{10}{1.7} + \frac{10}{7.3} + \frac{40.5}{6}} = 1.79\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1.5\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{6}{1.5\,\text{mA}} = \mathbf{4\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{1.5\,\text{mA}} = \mathbf{667\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 16.7\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 40.5$ with $R_{C1}\|R_{i2} = 3.55\,\text{k}\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 71.1\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{71.1\,\Omega} - \frac{1}{667\,\Omega}} = \mathbf{79.5\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{1.7}{150\,\mu\text{A}} = \mathbf{11.3\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{7.3}{150\,\mu\text{A}} = \mathbf{48.7\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 11.3\,\text{k}\Omega\|48.7\,\text{k}\Omega\|8.86\,\text{k}\Omega = 4.512\,\text{k}\Omega \geq 4\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	1.5 mA	1.31 mA	3.102 V
Q_2	6.46 mA	5.21 mA	4.682 V

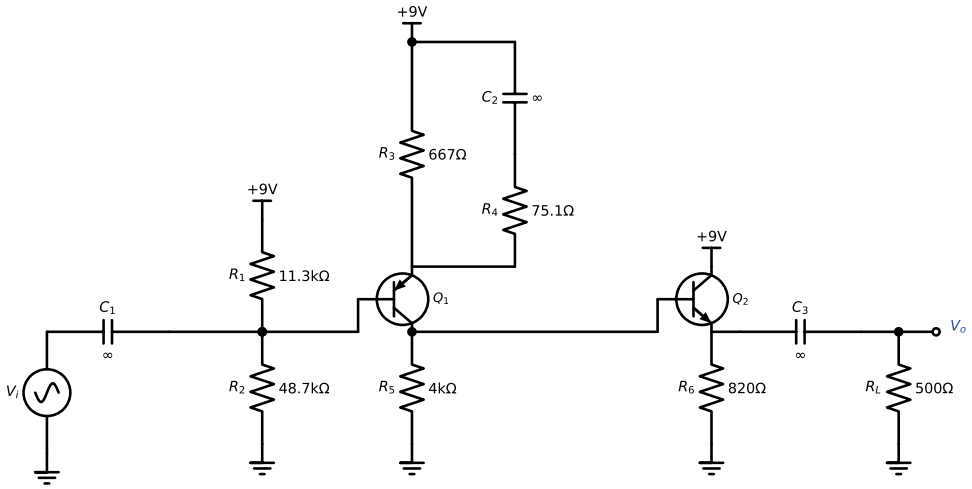
The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 38.42$ instead of 40 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 38.42/40 (exact $r_{e1} = 19.1\,\Omega$):

$$R_{E1}\|R_{G1} = (19.1\,\Omega + 71.1\,\Omega)\frac{38.42}{40} - 19.1\,\Omega = 67.5\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{75.1\,\Omega}$$

Re-analysis: $|A_v| = 40$ ✓.

- Input resistance (exact): $R_i = 4.48\,\text{k}\Omega \geq 4\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E2}(R_H\|R_L) = 1.62\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 2.858\,\text{V}$. The smallest, 1.62 V, is $\geq 1.5\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 1.7\,\% \leq 3\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

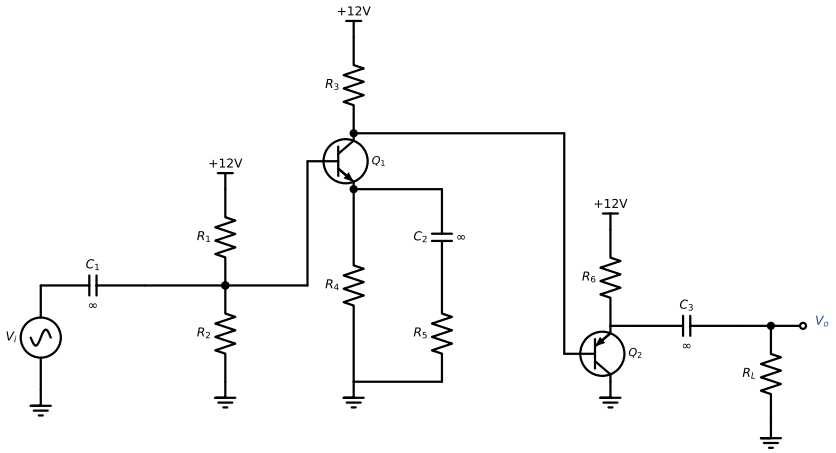
resistor	value
R_1 (R_{B1})	$11.3\,\mathrm{k\Omega}$
R_2 (R_{B2})	$48.7\,\mathrm{k\Omega}$
R_3 (R_{E1})	$667\,\Omega$
R_4 (R_{G1})	$75.1\,\Omega$
R_5 (R_{C1})	$4\,\mathrm{k\Omega}$
R_6 (R_H)	$820\,\Omega$



Answer

$R_1 = 11.3\,\mathrm{k\Omega}$, $R_2 = 48.7\,\mathrm{k\Omega}$, $R_3 = 667\,\Omega$, $R_4 = 75.1\,\Omega$, $R_5 = 4\,\mathrm{k\Omega}$, $R_6 = 820\,\Omega$

Solution 4E.8 **Medium**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{C1}$, $R_4 = R_{E1}$, $R_5 = R_{G1}$, $R_6 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 12 - 2.4 - 1 - 0.2 = 8.4\,\text{V} \;\geq\; \hat{V}_o = 2.4\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{8.4}{0.025} = 336 \geq 60$ ✓

Follower Q_2: its emitter sits at 0.7 V from the collector of Q_1, so $V_{RH} = V_{RC1} - 0.7 = 7.7\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC1} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC1} - 0.7 - \hat{V}_o}{\hat{V}_o} = 1\,\text{k}\Omega \cdot \frac{7.7 - 2.4}{2.4} = 2.21\,\text{k}\Omega$$

Take the E12 value just below: $R_H = \mathbf{2.2\,k\Omega}$. Then $I_{C2} \approx \frac{V_{RH}}{R_H} = \frac{7.7}{2.2\,\text{k}\Omega} = 3.5\,\text{mA}$, $r_{e2} = \frac{V_T}{I_{C2}} = 7.14\,\Omega$, and with $R_H \| R_L = 688\,\Omega$:

$$A_{v2} = \frac{R_H \| R_L}{r_{e2} + R_H \| R_L} = 0.9897, \qquad R_{i2} = (\beta+1)(r_{e2} + R_H \| R_L) = 70.2\,\text{k}\Omega$$

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 1.7\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 10.3\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E \| R_G) \approx \beta \frac{R_{C1}}{|A_{v1}|} = \beta \frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta/R_{i,spec}}{\frac{10}{10.3} + \frac{10}{1.7} + \frac{60.62}{8.4}} = 1.42\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1.2\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{8.4}{1.2\,\text{mA}} = \mathbf{7\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{1.2\,\text{mA}} = \mathbf{833\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 20.8\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 60.62$ with $R_{C1}\|R_{i2} = 6.36\,\text{k}\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 84.2\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{84.2\,\Omega} - \frac{1}{833\,\Omega}} = \mathbf{93.6\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{10.3}{120\,\mu\text{A}} = \mathbf{85.8\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{1.7}{120\,\mu\text{A}} = \mathbf{14.2\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1}+R_{E1}\|R_{G1}) = 85.8\,\text{k}\Omega\|14.2\,\text{k}\Omega\|10.6\,\text{k}\Omega = 5.664\,\text{k}\Omega \geq 5\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\|V_{CE}\|$
Q_1	1.2 mA	1.04 mA	4.06 V
Q_2	3.5 mA	2.87 mA	5.634 V

The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 57.5$ instead of 60 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 57.5/60 (exact $r_{e1} = 24.1\,\Omega$):

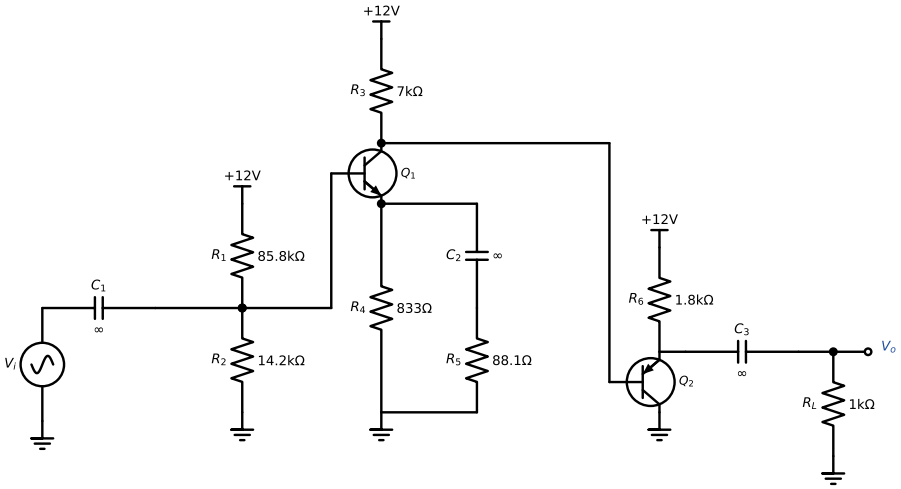
$$R_{E1}\|R_{G1} = (24.1\,\Omega + 84.2\,\Omega)\frac{57.5}{60} - 24.1\,\Omega = 79.7\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{88.1\,\Omega}$$

Re-analysis: $|A_v| = 60$ ✓.

The follower would clip at $\hat{V}_o = I_{E2}(R_H\|R_L) = 2.87\,\text{mA} \times 688\,\Omega = 1.97\,\text{V} < 2\,\text{V}$: its exact current is below the design value because the base currents shift the DC voltages along the direct-coupled chain. Adjust: lower R_H to the next E12 value, $R_H = \mathbf{1.8\,k\Omega}$ (more follower current), and analyse again. The exact gain is $|A_v| = 59.71$, within 0.5 % of the specification ✓.

- Input resistance (exact): $R_i = 5.63\,\text{k}\Omega \geq 5\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E2}(R_H\|R_L) = 2.391\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 3.862\,\text{V}$. The smallest, 2.391 V, is $\geq 2\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 1.7\,\% \leq 3\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

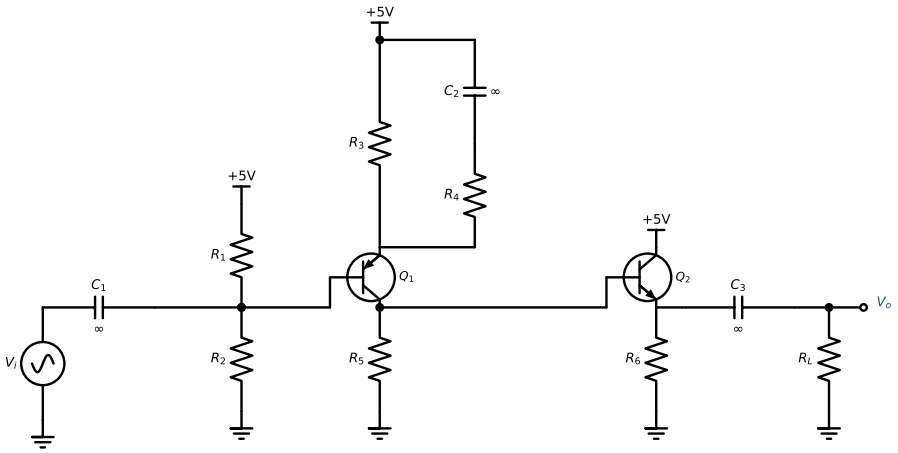
resistor	value
R_1 (R_{B1})	$85.8\,\mathrm{k\Omega}$
R_2 (R_{B2})	$14.2\,\mathrm{k\Omega}$
R_3 (R_{C1})	$7\,\mathrm{k\Omega}$
R_4 (R_{E1})	$833\,\Omega$
R_5 (R_{G1})	$88.1\,\Omega$
R_6 (R_H)	$1.8\,\mathrm{k\Omega}$



Answer

$R_1 = 85.8\,\mathrm{k\Omega}, \quad R_2 = 14.2\,\mathrm{k\Omega}, \quad R_3 = 7\,\mathrm{k\Omega}, \quad R_4 = 833\,\Omega, \quad R_5 = 88.1\,\Omega, \quad R_6 = 1.8\,\mathrm{k\Omega}$

Solution 4E.9 **Medium**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{E1}$, $R_4 = R_{G1}$, $R_5 = R_{C1}$, $R_6 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 5 - 1.2 - 1 - 0.2 = 2.6\,\text{V} \;\geq\; \hat{V}_o = 1.2\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{2.6}{0.025} = 104 \geq 30$ ✓

Follower Q_2: its emitter sits at 0.7 V from the collector of Q_1, so $V_{RH} = V_{RC1} - 0.7 = 1.9\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC1} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC1} - 0.7 - \hat{V}_o}{\hat{V}_o} = 200\,\Omega \cdot \frac{1.9 - 1.2}{1.2} = 117\,\Omega$$

Take the E12 value just below: $R_H = \mathbf{100\,\Omega}$. Then $I_{C2} \approx \frac{V_{RH}}{R_H} = \frac{1.9}{100\,\Omega} = 19\,\text{mA}$, $r_{e2} = \frac{V_T}{I_{C2}} = 1.32\,\Omega$, and with $R_H \| R_L = 66.7\,\Omega$:

$$A_{v2} = \frac{R_H \| R_L}{r_{e2} + R_H \| R_L} = 0.9806, \qquad R_{i2} = (\beta+1)(r_{e2} + R_H \| R_L) = 6.87\,\text{k}\Omega$$

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 3.3\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 1.7\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E \| R_G) \approx \beta \frac{R_{C1}}{|A_{v1}|} = \beta \frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta / I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta / R_{i,spec}}{\frac{10}{1.7} + \frac{10}{3.3} + \frac{30.59}{2.6}} = 3.22\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 2.7\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{2.6}{2.7\,\text{mA}} = \mathbf{963\,\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{2.7\,\text{mA}} = \mathbf{370\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 9.26\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 30.59$ with $R_{C1}\|R_{i2} = 845\,\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 18.3\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{18.3\,\Omega} - \frac{1}{370\,\Omega}} = \mathbf{19.3\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{1.7}{270\,\mu\text{A}} = \mathbf{6.3\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{3.3}{270\,\mu\text{A}} = \mathbf{12.2\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 6.3\,\text{k}\Omega\|12.2\,\text{k}\Omega\|2.79\,\text{k}\Omega = 1.669\,\text{k}\Omega \geq 1.5\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	2.7 mA	2.41 mA	1.924 V
Q_2	19 mA	14.6 mA	3.524 V

The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 28.39$ instead of 30 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 28.39/30 (exact $r_{e1} = 10.4\,\Omega$):

$$R_{E1}\|R_{G1} = (10.4\,\Omega + 18.3\,\Omega)\frac{28.39}{30} - 10.4\,\Omega = 16.8\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{17.6\,\Omega}$$

Re-analysis: $|A_v| = 30$ ✓.

The follower would clip at $\hat{V}_o = I_{E2}(R_H\|R_L) = 14.6\,\text{mA} \times 66.7\,\Omega = 0.9743\,\text{V} < 1\,\text{V}$: its exact current is below the design value because the base currents shift the DC voltages along the direct-coupled chain. Adjust: lower R_H to the next E12 value, $R_H = \mathbf{82\,\Omega}$ (more follower current), and analyse again. The exact gain is $|A_v| = 29.5$ instead of 30 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 29.5/30 (exact $r_{e1} = 10.4\,\Omega$):

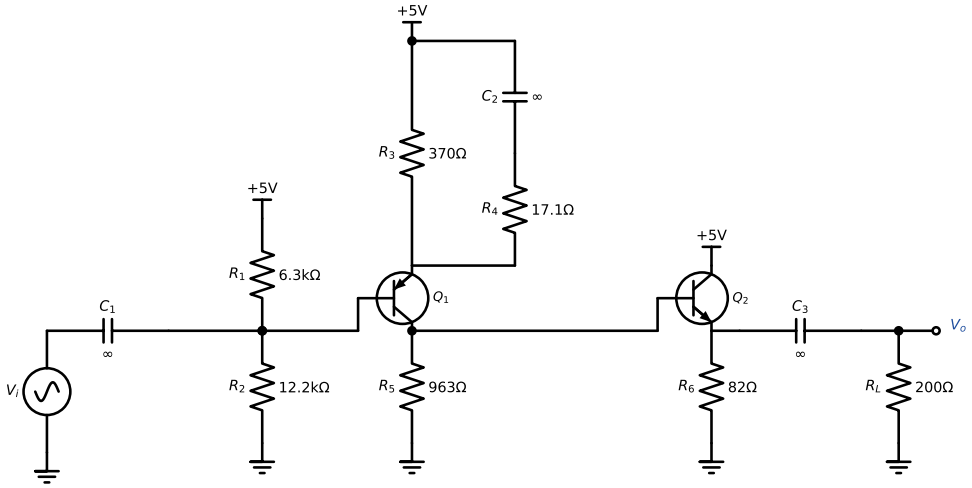
$$R_{E1}\|R_{G1} = (10.4\,\Omega + 16.8\,\Omega)\frac{29.5}{30} - 10.4\,\Omega = 16.3\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{17.1\,\Omega}$$

Re-analysis: $|A_v| = 30$ ✓.

- Input resistance (exact): $R_i = 1.64\,\text{k}\Omega \geq 1.5\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E2}(R_H\|R_L) = 1.166\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 1.715\,\text{V}$. The smallest, 1.166 V, is $\geq 1\,\text{V}$ ✓

- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 4.4\,\% \le 5\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \ge 1\,\text{V}$ ✓

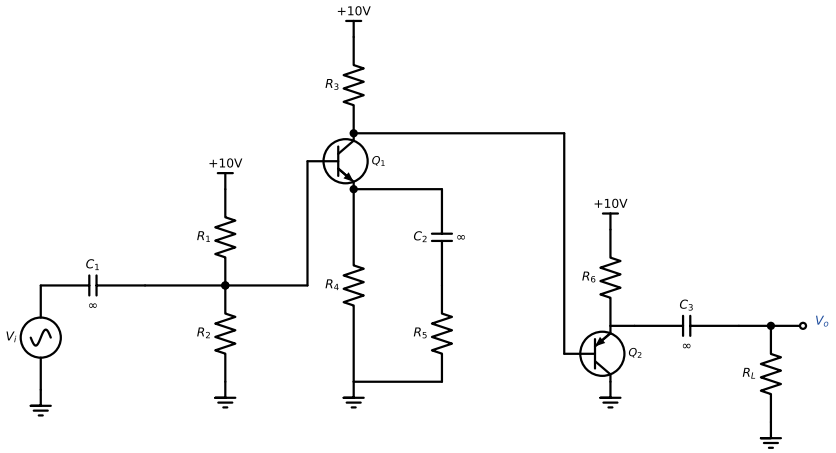
resistor	value
R_1 (R_{B1})	$6.3\,\text{k}\Omega$
R_2 (R_{B2})	$12.2\,\text{k}\Omega$
R_3 (R_{E1})	$370\,\Omega$
R_4 (R_{G1})	$17.1\,\Omega$
R_5 (R_{C1})	$963\,\Omega$
R_6 (R_H)	$82\,\Omega$



Answer

$R_1 = 6.3\,\text{k}\Omega$, $R_2 = 12.2\,\text{k}\Omega$, $R_3 = 370\,\Omega$, $R_4 = 17.1\,\Omega$, $R_5 = 963\,\Omega$, $R_6 = 82\,\Omega$

Solution 4E.10 **Medium**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{C1}$, $R_4 = R_{E1}$, $R_5 = R_{G1}$, $R_6 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 10 - 1.8 - 1 - 0.2 = 7\,\text{V} \;\geq\; \hat{V}_o = 1.8\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{7}{0.025} = 280 \geq 100$ ✓

Follower Q_2: its emitter sits at 0.7 V from the collector of Q_1, so $V_{RH} = V_{RC1} - 0.7 = 6.3\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC1} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC1} - 0.7 - \hat{V}_o}{\hat{V}_o} = 300\,\Omega \cdot \frac{6.3 - 1.8}{1.8} = 750\,\Omega$$

Take the E12 value just below: $R_H = \mathbf{680\,\Omega}$. Then $I_{C2} \approx \frac{V_{RH}}{R_H} = \frac{6.3}{680\,\Omega} = 9.26\,\text{mA}$, $r_{e2} = \frac{V_T}{I_{C2}} = 2.7\,\Omega$, and with $R_H \| R_L = 208\,\Omega$:

$$A_{v2} = \frac{R_H \| R_L}{r_{e2} + R_H \| R_L} = 0.9872, \qquad R_{i2} = (\beta+1)(r_{e2} + R_H \| R_L) = 21.3\,\text{k}\Omega$$

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 1.7\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 8.3\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E \| R_G) \approx \beta \frac{R_{C1}}{|A_{v1}|} = \beta \frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta / I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta / R_{i,spec}}{\frac{10}{8.3} + \frac{10}{1.7} + \frac{101.3}{7}} = 1.55\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1.5\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{7}{1.5\,\text{mA}} = \mathbf{4.67\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{1.5\,\text{mA}} = \mathbf{667\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 16.7\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 101.3$ with $R_{C1}\|R_{i2} = 3.83\,\text{k}\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 21.1\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{21.1\,\Omega} - \frac{1}{667\,\Omega}} = \mathbf{21.8\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{8.3}{150\,\mu\text{A}} = \mathbf{55.3\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{1.7}{150\,\mu\text{A}} = \mathbf{11.3\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 55.3\,\text{k}\Omega\|11.3\,\text{k}\Omega\|3.82\,\text{k}\Omega = 2.715\,\text{k}\Omega$
$< 3\,\text{k}\Omega$: the estimate above ignored that R_{C1} is loaded by the next stage (and used β for $\beta+1$). Take the next lower E12 value, $I_{C1} = 1.2\,\text{mA}$, and repeat:

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{7}{1.2\,\text{mA}} = \mathbf{5.83\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{1.2\,\text{mA}} = \mathbf{833\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 20.8\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 101.3$ with $R_{C1}\|R_{i2} = 4.58\,\text{k}\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 24.4\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{24.4\,\Omega} - \frac{1}{833\,\Omega}} = \mathbf{25.1\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{8.3}{120\,\mu\text{A}} = \mathbf{69.2\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{1.7}{120\,\mu\text{A}} = \mathbf{14.2\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 69.2\,\text{k}\Omega\|14.2\,\text{k}\Omega\|4.57\,\text{k}\Omega = 3.289\,\text{k}\Omega \geq 3\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	1.2 mA	1.04 mA	3.463 V
Q_2	9.26 mA	7.22 mA	5.04 V

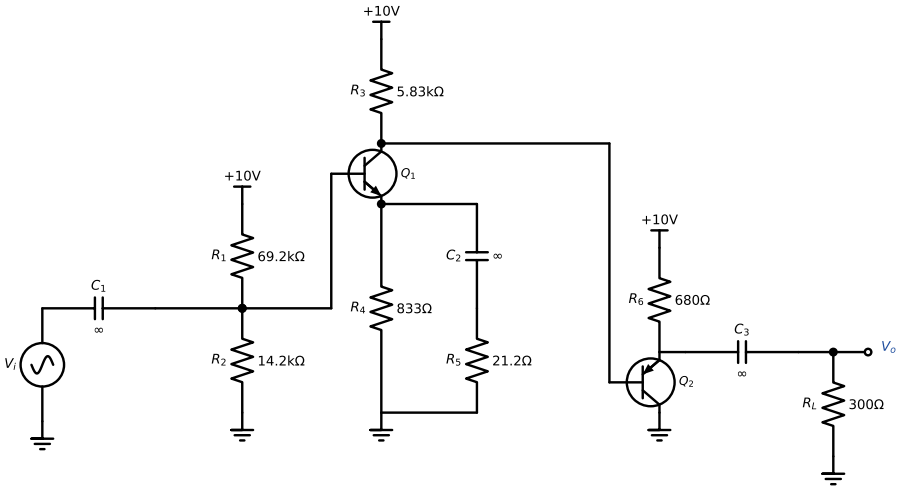
The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 92.3$ instead of 100 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 92.3/100 (exact $r_{e1} = 24\,\Omega$):

$$R_{E1}\|R_{G1} = (24\,\Omega + 24.4\,\Omega)\frac{92.3}{100} - 24\,\Omega = 20.6\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{21.2\,\Omega}$$

Re-analysis: $|A_v| = 100$ ✓.

- Input resistance (exact): $R_i = 3.26\,\text{k}\Omega \geq 3\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E2}(R_H \| R_L) = 1.503\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 3.209\,\text{V}$. The smallest, $1.503\,\text{V}$, is $\geq 1.5\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e + R_E \| R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T |A_v|}{4V_{RC}^2}$): THD $\approx 3.7\,\% \leq 5\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

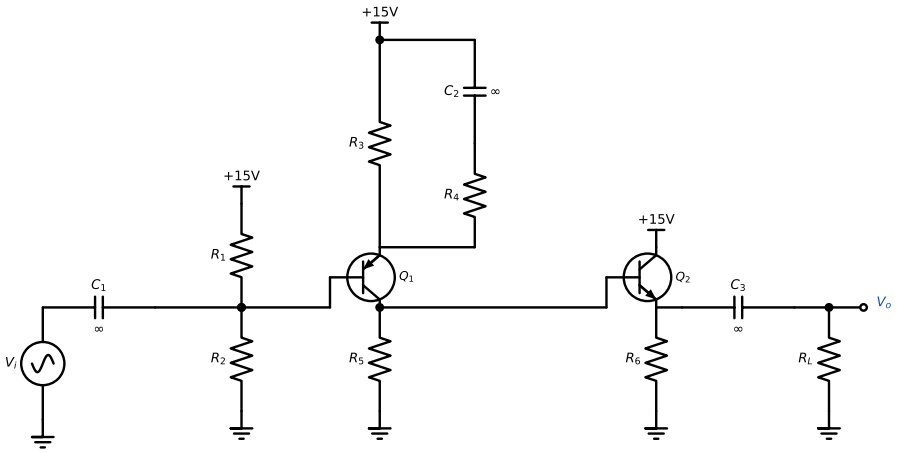
resistor	value
R_1 (R_{B1})	$69.2\,\text{k}\Omega$
R_2 (R_{B2})	$14.2\,\text{k}\Omega$
R_3 (R_{C1})	$5.83\,\text{k}\Omega$
R_4 (R_{E1})	$833\,\Omega$
R_5 (R_{G1})	$21.2\,\Omega$
R_6 (R_H)	$680\,\Omega$



Answer

$R_1 = 69.2\,\text{k}\Omega$, $R_2 = 14.2\,\text{k}\Omega$, $R_3 = 5.83\,\text{k}\Omega$, $R_4 = 833\,\Omega$, $R_5 = 21.2\,\Omega$, $R_6 = 680\,\Omega$

Solution 4E.11 **Medium**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{E1}$, $R_4 = R_{G1}$, $R_5 = R_{C1}$, $R_6 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e+Z_E}$, base resistance $(\beta+1)(r_e+Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 15 - 3.5 - 1 - 0.2 = 10.3\,\text{V} \;\geq\; \hat{V}_o = 3.5\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{10.3}{0.025} = 412 \geq 30$ ✓

Follower Q_2: its emitter sits at 0.7 V from the collector of Q_1, so $V_{RH} = V_{RC1} - 0.7 = 9.6\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC1} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC1} - 0.7 - \hat{V}_o}{\hat{V}_o} = 1\,\text{k}\Omega \cdot \frac{9.6 - 3.5}{3.5} = 1.74\,\text{k}\Omega$$

Take the E12 value just below: $R_H = \mathbf{1.5\,k\Omega}$. Then $I_{C2} \approx \frac{V_{RH}}{R_H} = \frac{9.6}{1.5\,\text{k}\Omega} = 6.4\,\text{mA}$, $r_{e2} = \frac{V_T}{I_{C2}} = 3.91\,\Omega$, and with $R_H \| R_L = 600\,\Omega$:

$$A_{v2} = \frac{R_H \| R_L}{r_{e2} + R_H \| R_L} = 0.9935, \qquad R_{i2} = (\beta+1)(r_{e2} + R_H \| R_L) = 61\,\text{k}\Omega$$

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 13.3\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 1.7\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 20$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E \| R_G) \approx \beta \frac{R_{C1}}{|A_{v1}|} = \beta \frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta/R_{i,spec}}{\frac{20}{1.7} + \frac{20}{13.3} + \frac{30.2}{10.3}} = 617\,\mu\text{A}$$

Take the E12 value just below: $I_{C1} = 560\,\mu\text{A}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{10.3}{560\,\mu\text{A}} = \mathbf{18.4\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{560\,\mu\text{A}} = \mathbf{1.79\,k\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 44.6\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 30.2$ with $R_{C1}\|R_{i2} = 14.1\,\text{k}\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 423\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{423\,\Omega} - \frac{1}{1.79\,\text{k}\Omega}} = \mathbf{555\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{1.7}{112\,\mu\text{A}} = \mathbf{15.2\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{13.3}{112\,\mu\text{A}} = \mathbf{119\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 15.2\,\text{k}\Omega\|119\,\text{k}\Omega\|47.3\,\text{k}\Omega = 10.48\,\text{k}\Omega \geq 10\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	$560\,\mu\text{A}$	$516\,\mu\text{A}$	$5.531\,\text{V}$
Q_2	$6.4\,\text{mA}$	$5.17\,\text{mA}$	$7.162\,\text{V}$

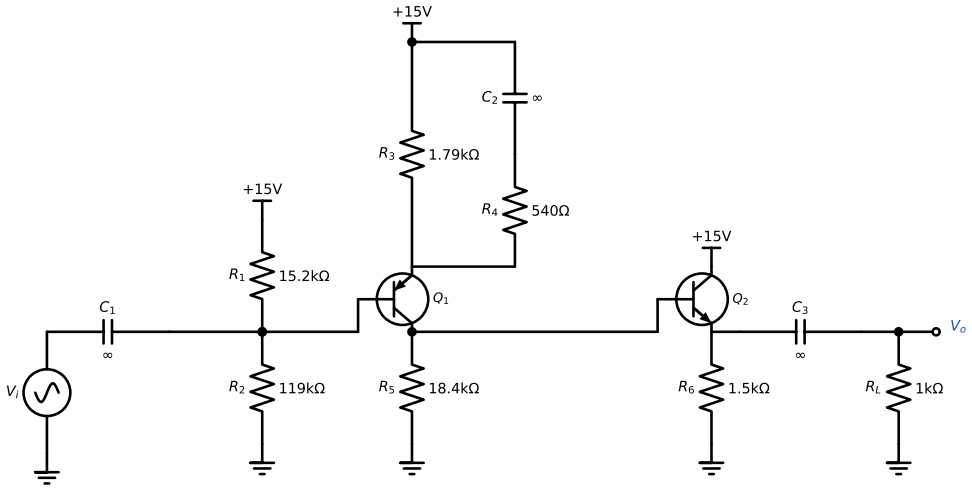
The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 29.43$ instead of 30 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 29.43/30 (exact $r_{e1} = 48.5\,\Omega$):

$$R_{E1}\|R_{G1} = (48.5\,\Omega + 423\,\Omega)\frac{29.43}{30} - 48.5\,\Omega = 414\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{540\,\Omega}$$

Re-analysis: $|A_v| = 30$ ✓.

- Input resistance (exact): $R_i = 10.4\,\text{k}\Omega \geq 10\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E2}(R_H\|R_L) = 3.104\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 5.289\,\text{V}$. The smallest, $3.104\,\text{V}$, is $\geq 3\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 1.1\,\% \leq 3\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

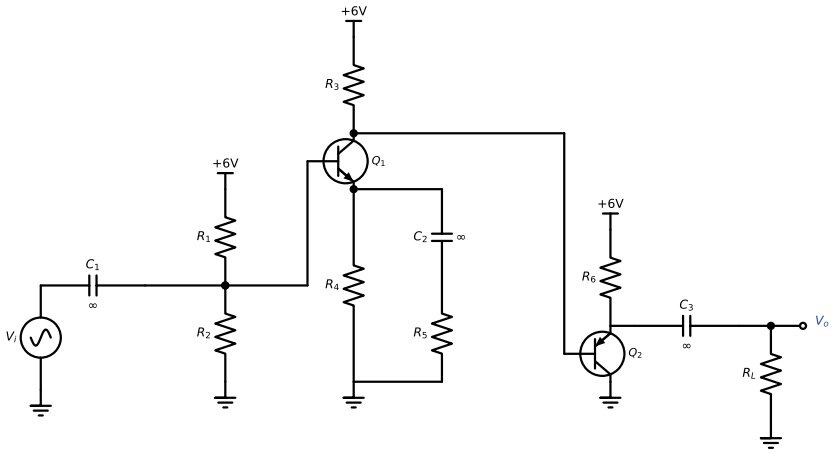
resistor	value
R_1 (R_{B1})	$15.2\,\mathrm{k\Omega}$
R_2 (R_{B2})	$119\,\mathrm{k\Omega}$
R_3 (R_{E1})	$1.79\,\mathrm{k\Omega}$
R_4 (R_{G1})	$540\,\Omega$
R_5 (R_{C1})	$18.4\,\mathrm{k\Omega}$
R_6 (R_H)	$1.5\,\mathrm{k\Omega}$



Answer

$R_1 = 15.2\,\mathrm{k\Omega}, \quad R_2 = 119\,\mathrm{k\Omega}, \quad R_3 = 1.79\,\mathrm{k\Omega}, \quad R_4 = 540\,\Omega, \quad R_5 = 18.4\,\mathrm{k\Omega}, \quad R_6 = 1.5\,\mathrm{k\Omega}$

Solution 4E.12 **Medium**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{C1}$, $R_4 = R_{E1}$, $R_5 = R_{G1}$, $R_6 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE1} \geq 1\,\text{V}$; use $V_{RE1} = 1\,\text{V}$.

To avoid clipping at saturation the collector must be able to swing down by $\hat{V}_o$ without V_{CE} dropping below $V_{CE,sat}$, and at cutoff V_{RC} must be at least $\hat{V}_o$: $V_{CC} - \hat{V}_o - V_{RE1} - V_{CE,sat} \geq V_{RC1} \geq \hat{V}_o$. Take the largest value (most gain, best linearity):

$$V_{RC1} = 6 - 1.2 - 1 - 0.2 = 3.6\,\text{V} \;\geq\; \hat{V}_o = 1.2\,\text{V} \;\checkmark$$

Maximum available gain ($R_G = 0$, R_C only): $|A_v|_{max} = \frac{R_C}{r_e} = \frac{V_{RC1}/I_C}{V_T/I_C} = \frac{V_{RC1}}{V_T} = \frac{3.6}{0.025} = 144 \geq 25$ ✓

Follower Q_2: its emitter sits at 0.7 V from the collector of Q_1, so $V_{RH} = V_{RC1} - 0.7 = 2.9\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC1} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC1} - 0.7 - \hat{V}_o}{\hat{V}_o} = 100\,\Omega \cdot \frac{2.9 - 1.2}{1.2} = 142\,\Omega$$

Take the E12 value just below: $R_H = \mathbf{120\,\Omega}$. Then $I_{C2} \approx \frac{V_{RH}}{R_H} = \frac{2.9}{120\,\Omega} = 24.2\,\text{mA}$, $r_{e2} = \frac{V_T}{I_{C2}} = 1.03\,\Omega$, and with $R_H \| R_L = 54.5\,\Omega$:

$$A_{v2} = \frac{R_H \| R_L}{r_{e2} + R_H \| R_L} = 0.9814, \qquad R_{i2} = (\beta+1)(r_{e2} + R_H \| R_L) = 5.61\,\text{k}\Omega$$

Step 3 — Bias current and resistors

Divider: $V_{RB2} = V_{B1} = 1.7\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 4.3\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E \| R_G) \approx \beta \frac{R_{C1}}{|A_{v1}|} = \beta \frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta / I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \geq R_{i,spec} \;\Rightarrow\; I_{C1} \leq \frac{\beta / R_{i,spec}}{\frac{10}{4.3} + \frac{10}{1.7} + \frac{25.47}{3.6}} = 3.27\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 2.7\,\text{mA}$.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{3.6}{2.7\,\text{mA}} = \mathbf{1.33\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{2.7\,\text{mA}} = \mathbf{370\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 9.26\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 25.47$ with $R_{C1}\|R_{i2} = 1.08\,\text{k}\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 33\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{33\,\Omega} - \frac{1}{370\,\Omega}} = \mathbf{36.3\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{4.3}{270\,\mu\text{A}} = \mathbf{15.9\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{1.7}{270\,\mu\text{A}} = \mathbf{6.3\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 15.9\,\text{k}\Omega\|6.3\,\text{k}\Omega\|4.27\,\text{k}\Omega = 2.194\,\text{k}\Omega \geq 2\,\text{k}\Omega$ ✓.

Step 4 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	2.7 mA	2.39 mA	2.173 V
Q_2	24.2 mA	18.4 mA	3.765 V

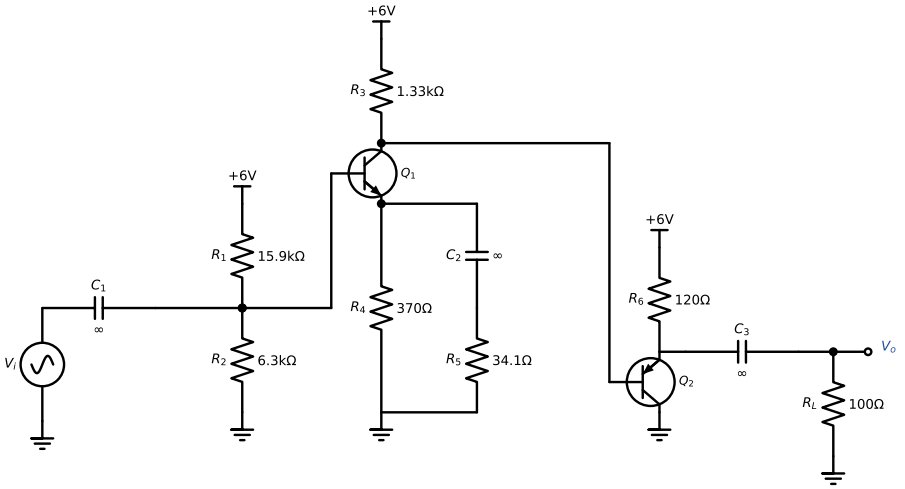
The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 23.95$ instead of 25 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 23.95/25 (exact $r_{e1} = 10.5\,\Omega$):

$$R_{E1}\|R_{G1} = (10.5\,\Omega + 33\,\Omega)\frac{23.95}{25} - 10.5\,\Omega = 31.2\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{34.1\,\Omega}$$

Re-analysis: $|A_v| = 25$ ✓.

- Input resistance (exact): $R_i = 2.18\,\text{k}\Omega \geq 2\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E2}(R_H\|R_L) = 1.006\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 1.925\,\text{V}$. The smallest, 1.006 V, is $\geq 1\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 2.3\,\% \leq 3\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V} \geq 1\,\text{V}$ ✓

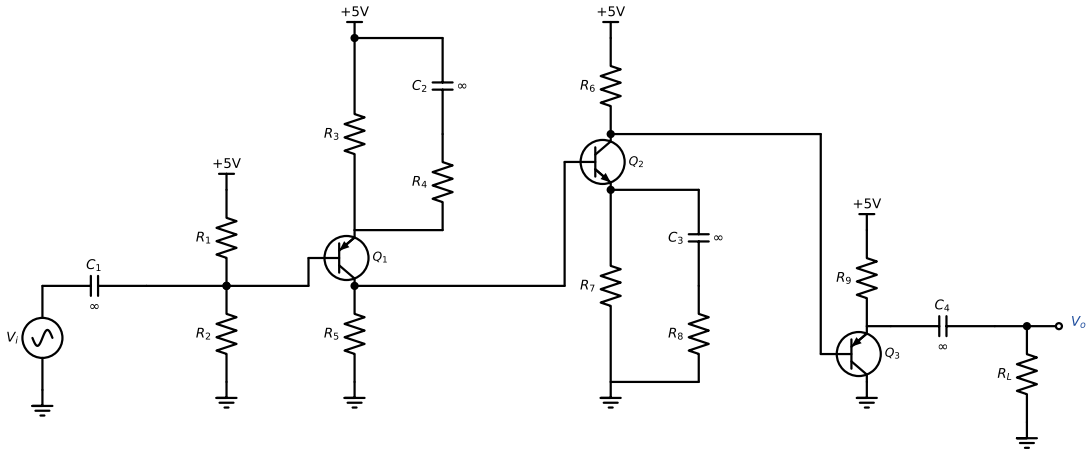
resistor	value
R_1 (R_{B1})	$15.9\,\mathrm{k\Omega}$
R_2 (R_{B2})	$6.3\,\mathrm{k\Omega}$
R_3 (R_{C1})	$1.33\,\mathrm{k\Omega}$
R_4 (R_{E1})	$370\,\Omega$
R_5 (R_{G1})	$34.1\,\Omega$
R_6 (R_H)	$120\,\Omega$



Answer

$R_1 = 15.9\,\mathrm{k\Omega}, \quad R_2 = 6.3\,\mathrm{k\Omega}, \quad R_3 = 1.33\,\mathrm{k\Omega}, \quad R_4 = 370\,\Omega, \quad R_5 = 34.1\,\Omega, \quad R_6 = 120\,\Omega$

Solution 4E.13 **Hard**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{E1}$, $R_4 = R_{G1}$, $R_5 = R_{C1}$, $R_6 = R_{C2}$, $R_7 = R_{E2}$, $R_8 = R_{G2}$, $R_9 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE2} \geq 1\,\text{V}$; use $V_{RE2} = 1\,\text{V}$.

Collector swing of Q_2: about $\hat{V}_o$ (the follower has a gain close to 1). Largest V_{RC2} without clipping:

$$V_{RC2} = V_{CC} - \hat{V}_o - V_{RE2} - V_{CE,sat} = 5 - 1.3 - 1 - 0.2 = 2.5\,\text{V}$$

Follower Q_3: its emitter sits at 0.7 V from the collector of Q_2, so $V_{RH} = V_{RC2} - 0.7 = 1.8\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC2} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC2} - 0.7 - \hat{V}_o}{\hat{V}_o} = 150\,\Omega \cdot \frac{1.8 - 1.3}{1.3} = 57.7\,\Omega$$

Use $R_H = \mathbf{50\,\Omega}$ (given). Then $I_{C3} \approx \frac{V_{RH}}{R_H} = \frac{1.8}{50\,\Omega} = 36\,\text{mA}$, $r_{e3} = \frac{V_T}{I_{C3}} = 694\,\text{m}\Omega$, and with $R_H \| R_L = 37.5\,\Omega$:

$$A_{v3} = \frac{R_H \| R_L}{r_{e3} + R_H \| R_L} = 0.9818, \qquad R_{i3} = (\beta+1)(r_{e3} + R_H \| R_L) = 3.86\,\text{k}\Omega$$

Step 3 — Stage 2 (common emitter, Q_2)

β-insensitive: $I_{C2} \geq N\frac{I_{C3}}{\beta} = 20 \cdot \frac{36\,\text{mA}}{100} = 7.2\,\text{mA}$; use $I_{C2} = 8\,\text{mA}$ (given).

$$R_{C2} = \frac{V_{RC2}}{I_{C2}} = \frac{2.5}{8\,\text{mA}} = \mathbf{312\,\Omega}, \quad R_{E2} = \frac{V_{RE2}}{I_{C2}} = \frac{1}{8\,\text{mA}} = \mathbf{125\,\Omega}, \quad r_{e2} = \frac{V_T}{I_{C2}} = 3.12\,\Omega$$

Gain $|A_{v2}| = \frac{R_{C2} \| R_{i3}}{r_{e2} + R_{E2} \| R_{G2}} = 6$ with $R_{C2} \| R_{i3} = 289\,\Omega$:

$$R_{E2} \| R_{G2} = \frac{R_{C2} \| R_{i3}}{|A_{v2}|} - r_{e2} = 45.1\,\Omega \;\Rightarrow\; R_{G2} = \frac{1}{\frac{1}{45.1\,\Omega} - \frac{1}{125\,\Omega}} = \mathbf{70.4\,\Omega}$$

Its input resistance: $R_{i2} = (\beta+1)(r_{e2} + R_{E2}\|R_{G2}) = 4.87\,\text{k}\Omega$.

Step 4 — Input stage (Q_1)

Gain left for stage 1: $|A_{v1}| = \frac{|A_v|}{|A_{v2}|A_{v3}} = \frac{100}{5.891} = 16.98$. The base of Q_2 fixes the collector of Q_1: $V_{RC1} = 1.7\,\text{V}$. Its collector swing is only $\hat{V}_o/(\text{gain after it}) = 0.2207\,\text{V}$, so $V_{RE1} \le V_{CC} - V_{RC1} - V_{CE,sat} - 0.221 = 2.879\,\text{V}$; with $V_{RE1} \ge 1\,\text{V}$ use $V_{RE1} = 1.8\,\text{V}$.

Divider: $V_{RB2} = V_{B1} = 2.5\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 2.5\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 20$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \ge R_{i,spec} \;\Rightarrow\; I_{C1} \le \frac{\beta/R_{i,spec}}{\frac{20}{2.5} + \frac{20}{2.5} + \frac{16.98}{1.7}} = 1.92\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1.8\,\text{mA}$.

Check $I_{C1} \ge N\frac{I_{C2}}{\beta} = 1.6\,\text{mA}$ ✓.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{1.7}{1.8\,\text{mA}} = \mathbf{944\,\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1.8}{1.8\,\text{mA}} = \mathbf{1\,\text{k}\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 13.9\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 16.98$ with $R_{C1}\|R_{i2} = 791\,\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 32.7\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{32.7\,\Omega} - \frac{1}{1\,\text{k}\Omega}} = \mathbf{33.8\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{2.5}{360\,\mu\text{A}} = \mathbf{6.94\,\text{k}\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{2.5}{360\,\mu\text{A}} = \mathbf{6.94\,\text{k}\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 6.94\,\text{k}\Omega\|6.94\,\text{k}\Omega\|4.71\,\text{k}\Omega = 1.998\,\text{k}\Omega$

$< 2\,\text{k}\Omega$: the estimate above ignored that R_{C1} is loaded by the next stage (and used β for $\beta+1$). Take the next lower E12 value, $I_{C1} = 1.5\,\text{mA}$, and repeat:

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{1.7}{1.5\,\text{mA}} = \mathbf{1.13\,\text{k}\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1.8}{1.5\,\text{mA}} = \mathbf{1.2\,\text{k}\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 16.7\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 16.98$ with $R_{C1}\|R_{i2} = 919\,\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 37.5\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{37.5\,\Omega} - \frac{1}{1.2\,\text{k}\Omega}} = \mathbf{38.7\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{2.5}{300\,\mu\text{A}} = \mathbf{8.33\,\text{k}\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{2.5}{300\,\mu\text{A}} = \mathbf{8.33\,\text{k}\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 8.33\,\text{k}\Omega\|8.33\,\text{k}\Omega\|5.47\,\text{k}\Omega = 2.365\,\text{k}\Omega \ge 2\,\text{k}\Omega$ ✓.

Step 5 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\lvert V_{CE}\rvert$
Q_1	1.5 mA	1.44 mA	1.709 V
Q_2	8 mA	6.74 mA	2.125 V
Q_3	36 mA	26.2 mA	3.676 V

The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 95.16$ instead of 100 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 95.16/100 (exact $r_{e1} = 17.4\,\Omega$):

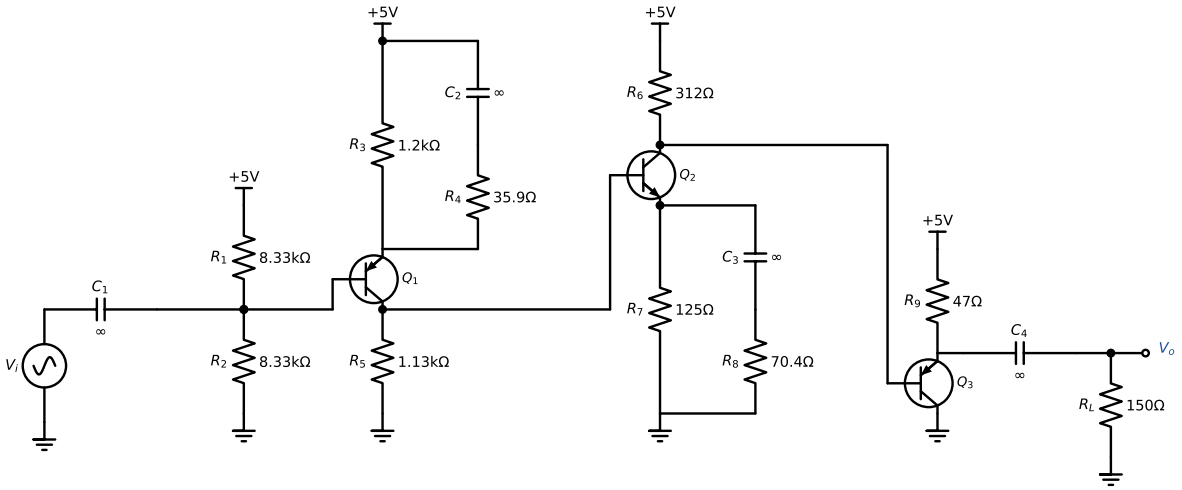
$$R_{E1}\|R_{G1} = (17.4\,\Omega + 37.5\,\Omega)\frac{95.16}{100} - 17.4\,\Omega = 34.8\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{35.9\,\Omega}$$

Re-analysis: $|A_v| = 100$ ✓.

The follower would clip at $\hat{V}_o = I_{E3}(R_H\|R_L) = 26.2\,\text{mA} \times 37.5\,\Omega = 0.9833\,\text{V} < 1\,\text{V}$: its exact current is below the design value because the base currents shift the DC voltages along the direct-coupled chain. Adjust: lower R_H to the next E12 value, $R_H = \mathbf{47\,\Omega}$ (more follower current), and analyse again. The exact gain is $|A_v| = 99.67$, within 0.5 % of the specification ✓.

- Input resistance (exact): $R_i = 2.33\,\text{k}\Omega \geq 2\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E3}(R_H\|R_L) = 1.042\,\text{V}$; Q_2 clips (saturation or cutoff) at $\hat{V}_o = 1.885\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 7.585\,\text{V}$. The smallest, 1.042 V, is $\geq 1\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 1 + 0.96\,\% = 2\,\% \leq 3\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1.8\,\text{V}$, $V_{RE2} = 1\,\text{V} \geq 1\,\text{V}$ ✓

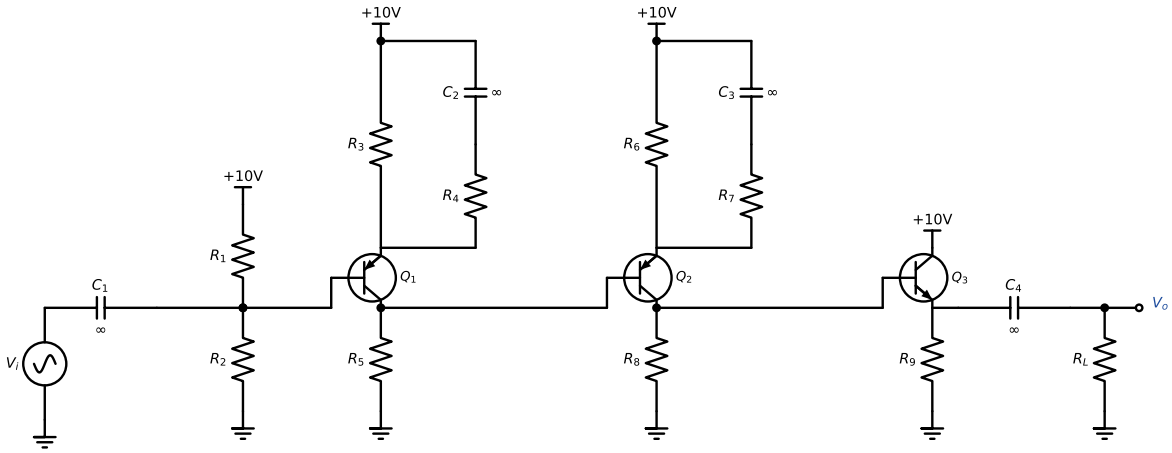
resistor	value
R_1 (R_{B1})	8.33 kΩ
R_2 (R_{B2})	8.33 kΩ
R_3 (R_{E1})	1.2 kΩ
R_4 (R_{G1})	35.9 Ω
R_5 (R_{C1})	1.13 kΩ
R_6 (R_{C2})	312 Ω
R_7 (R_{E2})	125 Ω
R_8 (R_{G2})	70.4 Ω
R_9 (R_H)	47 Ω



Answer

$R_1 = 8.33\,\text{k}\Omega, \quad R_2 = 8.33\,\text{k}\Omega, \quad R_3 = 1.2\,\text{k}\Omega, \quad R_4 = 35.9\,\Omega, \quad R_5 = 1.13\,\text{k}\Omega, \quad R_6 = 312\,\Omega, \quad R_7 = 125\,\Omega, \quad R_8 = 70.4\,\Omega, \quad R_9 = 47\,\Omega$

Solution 4E.14 **Hard**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{E1}$, $R_4 = R_{G1}$, $R_5 = R_{C1}$, $R_6 = R_{E2}$, $R_7 = R_{G2}$, $R_8 = R_{C2}$, $R_9 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE2} \geq 1\,\text{V}$; use $V_{RE2} = 1\,\text{V}$.

Collector swing of Q_2: about $\hat{V}_o$ (the follower has a gain close to 1). Largest V_{RC2} without clipping:

$$V_{RC2} = V_{CC} - \hat{V}_o - V_{RE2} - V_{CE,sat} = 10 - 2.4 - 1 - 0.2 = 6.4\,\text{V}$$

Follower Q_3: its emitter sits at 0.7 V from the collector of Q_2, so $V_{RH} = V_{RC2} - 0.7 = 5.7\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC2} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC2} - 0.7 - \hat{V}_o}{\hat{V}_o} = 500\,\Omega \cdot \frac{5.7 - 2.4}{2.4} = 687\,\Omega$$

Take the E12 value just below: $R_H = \mathbf{680\,\Omega}$. Then $I_{C3} \approx \frac{V_{RH}}{R_H} = \frac{5.7}{680\,\Omega} = 8.38\,\text{mA}$, $r_{e3} = \frac{V_T}{I_{C3}} = 2.98\,\Omega$, and with $R_H \| R_L = 288\,\Omega$:

$$A_{v3} = \frac{R_H \| R_L}{r_{e3} + R_H \| R_L} = 0.9898, \qquad R_{i3} = (\beta+1)(r_{e3} + R_H \| R_L) = 29.4\,\text{k}\Omega$$

Step 3 — Stage 2 (common emitter, Q_2)

β-insensitive: $I_{C2} \geq N \frac{I_{C3}}{\beta} = 10 \cdot \frac{8.38\,\text{mA}}{100} = 838\,\mu\text{A}$; take the E12 value just above, $I_{C2} = 1\,\text{mA}$.

$$R_{C2} = \frac{V_{RC2}}{I_{C2}} = \frac{6.4}{1\,\text{mA}} = \mathbf{6.4\,k\Omega}, \quad R_{E2} = \frac{V_{RE2}}{I_{C2}} = \frac{1}{1\,\text{mA}} = \mathbf{1\,k\Omega}, \quad r_{e2} = \frac{V_T}{I_{C2}} = 25\,\Omega$$

Gain $|A_{v2}| = \frac{R_{C2}\|R_{i3}}{r_{e2}+R_{E2}\|R_{G2}} = 10$ with $R_{C2}\|R_{i3} = 5.26\,\text{k}\Omega$:

$$R_{E2}\|R_{G2} = \frac{R_{C2}\|R_{i3}}{|A_{v2}|} - r_{e2} = 501\,\Omega \;\Rightarrow\; R_{G2} = \frac{1}{\frac{1}{501\,\Omega} - \frac{1}{1\,\text{k}\Omega}} = \mathbf{1\,k\Omega}$$

Its input resistance: $R_{i2} = (\beta+1)(r_{e2} + R_{E2}\|R_{G2}) = 53.1\,\text{k}\Omega$.

Step 4 — Input stage (Q_1)

Gain left for stage 1: $|A_{v1}| = \frac{|A_v|}{|A_{v2}|A_{v3}} = \frac{200}{9.898} = 20.21$. The base of Q_2 fixes the collector of Q_1: $V_{RC1} = 8.3\,\text{V}$. Its collector swing is only $\hat{V}_o/(\text{gain after it}) = 0.2425\,\text{V}$, so $V_{RE1} \le V_{CC} - V_{RC1} - V_{CE,sat} - 0.242 = 1.258\,\text{V}$; with $V_{RE1} \ge 1\,\text{V}$ use $V_{RE1} = 1\,\text{V}$.

Divider: $V_{RB2} = V_{B1} = 8.3\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 1.7\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \ge R_{i,spec} \;\Rightarrow\; I_{C1} \le \frac{\beta/R_{i,spec}}{\frac{10}{1.7} + \frac{10}{8.3} + \frac{20.21}{8.3}} = 2.1\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1.8\,\text{mA}$.

Check $I_{C1} \ge N\frac{I_{C2}}{\beta} = 100\,\mu\text{A}$ ✓.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{8.3}{1.8\,\text{mA}} = \mathbf{4.61\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{1.8\,\text{mA}} = \mathbf{556\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 13.9\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 20.21$ with $R_{C1}\|R_{i2} = 4.24\,\text{k}\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 196\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{196\,\Omega} - \frac{1}{556\,\Omega}} = \mathbf{303\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{1.7}{180\,\mu\text{A}} = \mathbf{9.44\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{8.3}{180\,\mu\text{A}} = \mathbf{46.1\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 9.44\,\text{k}\Omega\|46.1\,\text{k}\Omega\|21.2\,\text{k}\Omega = 5.723\,\text{k}\Omega \ge 5\,\text{k}\Omega$ ✓.

Step 5 — Check by analysis

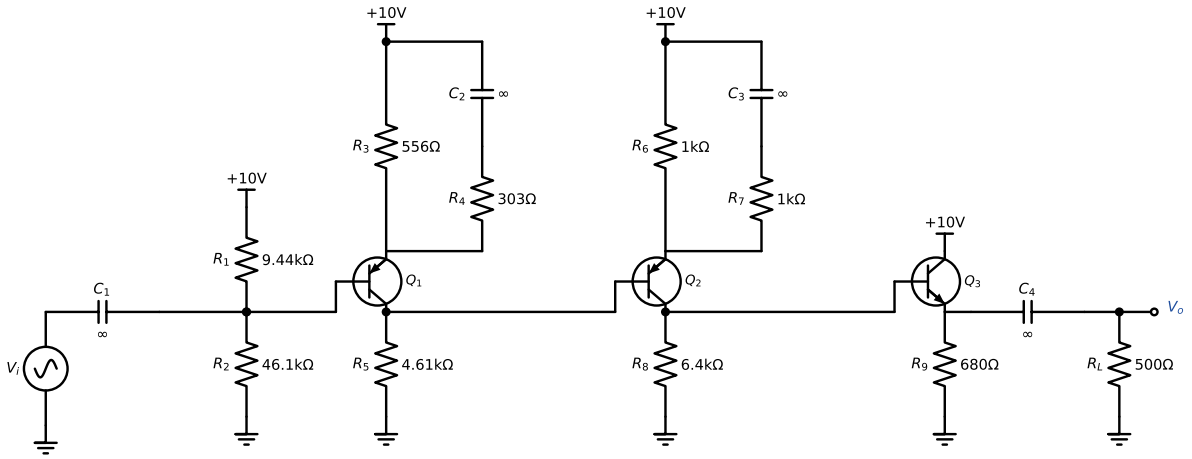
Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\|V_{CE}\|$
Q_1	1.8 mA	1.56 mA	1.821 V
Q_2	1 mA	1.98 mA	3.641 V
Q_3	8.38 mA	15.9 mA	0.9426 V

The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 199.2$, within 0.5 % of the specification ✓.

- Input resistance (exact): $R_i = 5.74\,\text{k}\Omega \geq 5\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E3}(R_H \| R_L) = 4.591\,\text{V}$; Q_2 clips (saturation or cutoff) at $\hat{V}_o = 3.422\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 16.33\,\text{V}$. The smallest, $3.422\,\text{V}$, is $\geq 2\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e + R_E \| R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T |A_v|}{4V_{RC}^2}$): THD $\approx 0.057 + 0.12\,\% = 0.18\,\% \leq 5\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V}$, $V_{RE2} = 1\,\text{V} \geq 1\,\text{V}$ ✓

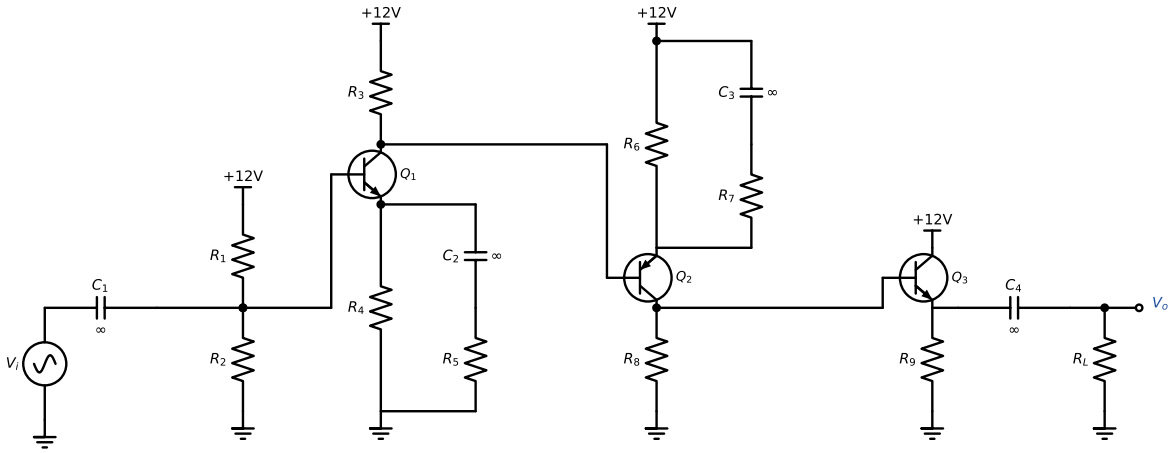
resistor	value
R_1 (R_{B1})	$9.44\,\text{k}\Omega$
R_2 (R_{B2})	$46.1\,\text{k}\Omega$
R_3 (R_{E1})	$556\,\Omega$
R_4 (R_{G1})	$303\,\Omega$
R_5 (R_{C1})	$4.61\,\text{k}\Omega$
R_6 (R_{E2})	$1\,\text{k}\Omega$
R_7 (R_{G2})	$1\,\text{k}\Omega$
R_8 (R_{C2})	$6.4\,\text{k}\Omega$
R_9 (R_H)	$680\,\Omega$



Answer

$R_1 = 9.44\,\text{k}\Omega$, $R_2 = 46.1\,\text{k}\Omega$, $R_3 = 556\,\Omega$, $R_4 = 303\,\Omega$, $R_5 = 4.61\,\text{k}\Omega$, $R_6 = 1\,\text{k}\Omega$, $R_7 = 1\,\text{k}\Omega$, $R_8 = 6.4\,\text{k}\Omega$, $R_9 = 680\,\Omega$

Solution 4E.15 **Hard**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{C1}$, $R_4 = R_{E1}$, $R_5 = R_{G1}$, $R_6 = R_{E2}$, $R_7 = R_{G2}$, $R_8 = R_{C2}$, $R_9 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE2} \geq 1\,\text{V}$; use $V_{RE2} = 1\,\text{V}$.

Collector swing of Q_2: about $\hat{V}_o$ (the follower has a gain close to 1). Largest V_{RC2} without clipping:

$$V_{RC2} = V_{CC} - \hat{V}_o - V_{RE2} - V_{CE,sat} = 12 - 2.4 - 1 - 0.2 = 8.4\,\text{V}$$

Follower Q_3: its emitter sits at 0.7 V from the collector of Q_2, so $V_{RH} = V_{RC2} - 0.7 = 7.7\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC2} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC2} - 0.7 - \hat{V}_o}{\hat{V}_o} = 1\,\text{k}\Omega \cdot \frac{7.7 - 2.4}{2.4} = 2.21\,\text{k}\Omega$$

Take the E12 value just below: $R_H = \mathbf{2.2\,k\Omega}$. Then $I_{C3} \approx \frac{V_{RH}}{R_H} = \frac{7.7}{2.2\,\text{k}\Omega} = 3.5\,\text{mA}$, $r_{e3} = \frac{V_T}{I_{C3}} = 7.14\,\Omega$, and with $R_H \| R_L = 688\,\Omega$:

$$A_{v3} = \frac{R_H \| R_L}{r_{e3} + R_H \| R_L} = 0.9897, \qquad R_{i3} = (\beta+1)(r_{e3} + R_H \| R_L) = 70.2\,\text{k}\Omega$$

Step 3 — Stage 2 (common emitter, Q_2)

β-insensitive: $I_{C2} \geq N \frac{I_{C3}}{\beta} = 20 \cdot \frac{3.5\,\text{mA}}{100} = 700\,\mu\text{A}$; take the E12 value just above, $I_{C2} = 820\,\mu\text{A}$.

$$R_{C2} = \frac{V_{RC2}}{I_{C2}} = \frac{8.4}{820\,\mu\text{A}} = \mathbf{10.2\,k\Omega}, \quad R_{E2} = \frac{V_{RE2}}{I_{C2}} = \frac{1}{820\,\mu\text{A}} = \mathbf{1.22\,k\Omega}, \quad r_{e2} = \frac{V_T}{I_{C2}} = 30.5\,\Omega$$

Gain $|A_{v2}| = \frac{R_{C2}\|R_{i3}}{r_{e2}+R_{E2}\|R_{G2}} = 8$ with $R_{C2}\|R_{i3} = 8.94\,\text{k}\Omega$:

$$R_{E2}\|R_{G2} = \frac{R_{C2}\|R_{i3}}{|A_{v2}|} - r_{e2} = 1.09\,\text{k}\Omega \;\Rightarrow\; R_{G2} = \frac{1}{\frac{1}{1.09\,\text{k}\Omega} - \frac{1}{1.22\,\text{k}\Omega}} = \mathbf{9.99\,k\Omega}$$

Its input resistance: $R_{i2} = (\beta+1)(r_{e2} + R_{E2}\|R_{G2}) = 113\,\text{k}\Omega$.

Step 4 — Input stage (Q_1)

Gain left for stage 1: $|A_{v1}| = \frac{|A_v|}{|A_{v2}|A_{v3}} = \frac{150}{7.918} = 18.94$. The base of Q_2 fixes the collector of Q_1: $V_{RC1} = 1.7\,\text{V}$. Its collector swing is only $\hat{V}_o$/(gain after it) $= 0.3031\,\text{V}$, so $V_{RE1} \le V_{CC} - V_{RC1} - V_{CE,sat} - 0.303 = 9.797\,\text{V}$; with $V_{RE1} \ge 1\,\text{V}$ use $V_{RE1} = 1.5\,\text{V}$.

Divider: $V_{RB2} = V_{B1} = 2.2\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 9.8\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 20$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \ge R_{i,spec} \;\Rightarrow\; I_{C1} \le \frac{\beta/R_{i,spec}}{\frac{20}{9.8} + \frac{20}{2.2} + \frac{18.94}{1.7}} = 1.12\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1\,\text{mA}$.

Check $I_{C1} \ge N\frac{I_{C2}}{\beta} = 164\,\mu\text{A}$ ✓.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{1.7}{1\,\text{mA}} = \mathbf{1.7\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1.5}{1\,\text{mA}} = \mathbf{1.5\,k\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 25\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 18.94$ with $R_{C1}\|R_{i2} = 1.67\,\text{k}\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 63.4\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{63.4\,\Omega} - \frac{1}{1.5\,\text{k}\Omega}} = \mathbf{66.2\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{9.8}{200\,\mu\text{A}} = \mathbf{49\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{2.2}{200\,\mu\text{A}} = \mathbf{11\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 49\,\text{k}\Omega\|11\,\text{k}\Omega\|8.93\,\text{k}\Omega = 4.478\,\text{k}\Omega \ge 4\,\text{k}\Omega$ ✓.

Step 5 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\|V_{CE}\|$
Q_1	1 mA	935 µA	9.007 V
Q_2	820 µA	712 µA	4.121 V
Q_3	3.5 mA	2.84 mA	5.698 V

The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 143.3$ instead of 150 (the hand design ignored α and the base currents). As the

slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 143.3/150 (exact $r_{e1} = 26.7\,\Omega$):

$$R_{E1}\|R_{G1} = (26.7\,\Omega + 63.4\,\Omega)\frac{143.3}{150} - 26.7\,\Omega = 59.4\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{61.8\,\Omega}$$

Re-analysis: $|A_v| = 150$ ✓.

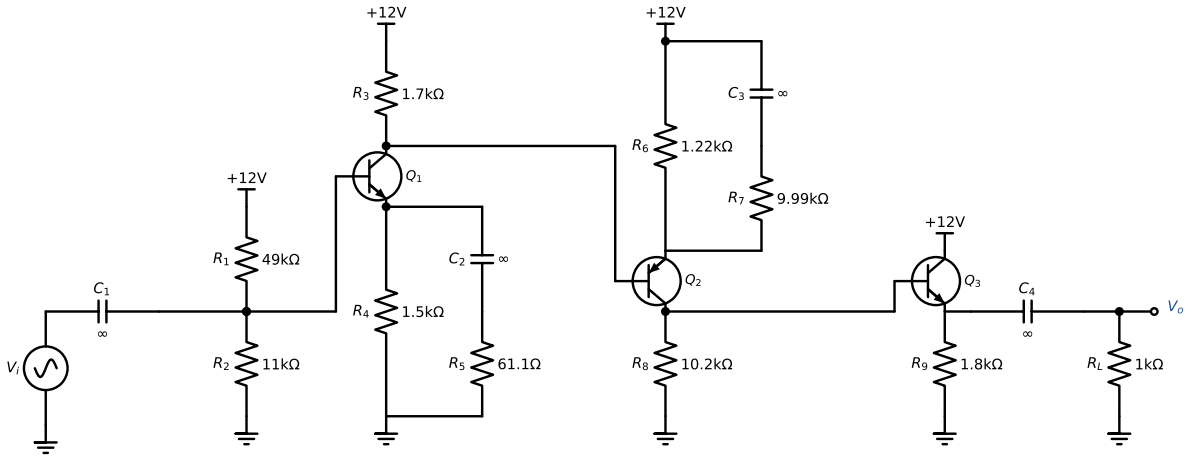
The follower would clip at $\hat{V}_o = I_{E3}(R_H\|R_L) = 2.84\,\text{mA} \times 688\,\Omega = 1.95\,\text{V} < 2\,\text{V}$: its exact current is below the design value because the base currents shift the DC voltages along the direct-coupled chain. Adjust: lower R_H to the next E12 value, $R_H = \mathbf{1.8\,k\Omega}$ (more follower current), and analyse again. The exact gain is $|A_v| = 148.9$ instead of 150 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 148.9/150 (exact $r_{e1} = 26.7\,\Omega$):

$$R_{E1}\|R_{G1} = (26.7\,\Omega + 59.4\,\Omega)\frac{148.9}{150} - 26.7\,\Omega = 58.7\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{61.1\,\Omega}$$

Re-analysis: $|A_v| = 150$ ✓.

- Input resistance (exact): $R_i = 4.4\,\text{k}\Omega \geq 4\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E3}(R_H\|R_L) = 2.36\,\text{V}$; Q_2 clips (saturation or cutoff) at $\hat{V}_o = 3.94\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 12.22\,\text{V}$. The smallest, 2.36 V, is $\geq 2\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e + R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T |A_v|}{4V_{RC}^2}$): THD $\approx 1.2 + 0.24\,\% = 1.4\,\% \leq 5\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1.5\,\text{V}$, $V_{RE2} = 1\,\text{V} \geq 1\,\text{V}$ ✓

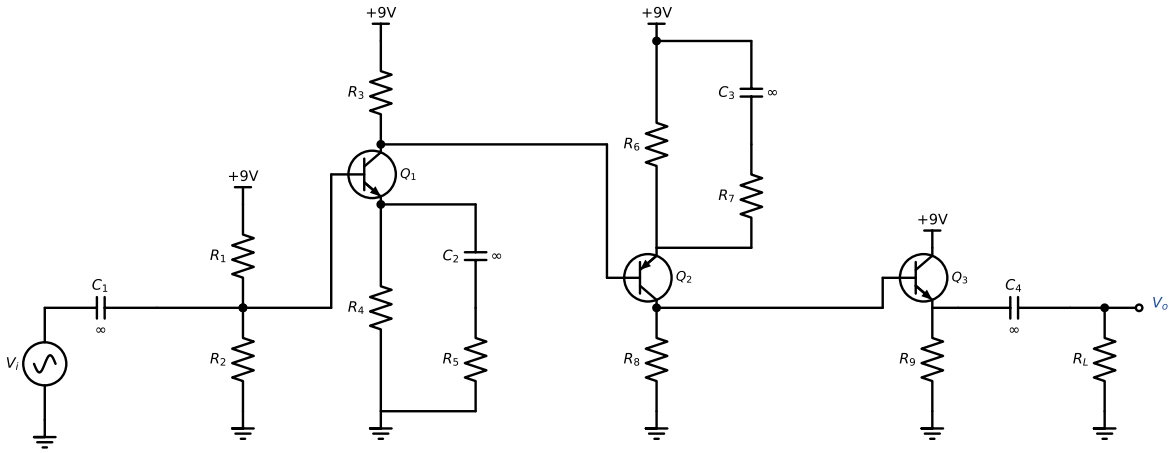
resistor	value
R_1 (R_{B1})	49 kΩ
R_2 (R_{B2})	11 kΩ
R_3 (R_{C1})	1.7 kΩ
R_4 (R_{E1})	1.5 kΩ
R_5 (R_{G1})	61.1 Ω
R_6 (R_{E2})	1.22 kΩ
R_7 (R_{G2})	9.99 kΩ
R_8 (R_{C2})	10.2 kΩ
R_9 (R_H)	1.8 kΩ



Answer

$R_1 = 49\,\text{k}\Omega$, $R_2 = 11\,\text{k}\Omega$, $R_3 = 1.7\,\text{k}\Omega$, $R_4 = 1.5\,\text{k}\Omega$, $R_5 = 61.1\,\Omega$, $R_6 = 1.22\,\text{k}\Omega$, $R_7 = 9.99\,\text{k}\Omega$, $R_8 = 10.2\,\text{k}\Omega$, $R_9 = 1.8\,\text{k}\Omega$

Solution 4E.16 **Hard**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{C1}$, $R_4 = R_{E1}$, $R_5 = R_{G1}$, $R_6 = R_{E2}$, $R_7 = R_{G2}$, $R_8 = R_{C2}$, $R_9 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE2} \geq 1\,\text{V}$; use $V_{RE2} = 1\,\text{V}$.

Collector swing of Q_2: about $\hat{V}_o$ (the follower has a gain close to 1). Largest V_{RC2} without clipping:

$$V_{RC2} = V_{CC} - \hat{V}_o - V_{RE2} - V_{CE,sat} = 9 - 1.8 - 1 - 0.2 = 6\,\text{V}$$

Follower Q_3: its emitter sits at 0.7 V from the collector of Q_2, so $V_{RH} = V_{RC2} - 0.7 = 5.3\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC2} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC2} - 0.7 - \hat{V}_o}{\hat{V}_o} = 300\,\Omega \cdot \frac{5.3 - 1.8}{1.8} = 583\,\Omega$$

Take the E12 value just below: $R_H = \mathbf{560\,\Omega}$. Then $I_{C3} \approx \frac{V_{RH}}{R_H} = \frac{5.3}{560\,\Omega} = 9.46\,\text{mA}$, $r_{e3} = \frac{V_T}{I_{C3}} = 2.64\,\Omega$, and with $R_H \| R_L = 195\,\Omega$:

$$A_{v3} = \frac{R_H \| R_L}{r_{e3} + R_H \| R_L} = 0.9867, \qquad R_{i3} = (\beta+1)(r_{e3} + R_H \| R_L) = 20\,\text{k}\Omega$$

Step 3 — Stage 2 (common emitter, Q_2)

β-insensitive: $I_{C2} \geq N \frac{I_{C3}}{\beta} = 10 \cdot \frac{9.46\,\text{mA}}{100} = 946\,\mu\text{A}$; take the E12 value just above, $I_{C2} = 1\,\text{mA}$.

$$R_{C2} = \frac{V_{RC2}}{I_{C2}} = \frac{6}{1\,\text{mA}} = \mathbf{6\,k\Omega}, \quad R_{E2} = \frac{V_{RE2}}{I_{C2}} = \frac{1}{1\,\text{mA}} = \mathbf{1\,k\Omega}, \quad r_{e2} = \frac{V_T}{I_{C2}} = 25\,\Omega$$

Gain $|A_{v2}| = \frac{R_{C2}\|R_{i3}}{r_{e2}+R_{E2}\|R_{G2}} = 8$ with $R_{C2}\|R_{i3} = 4.62\,\text{k}\Omega$:

$$R_{E2}\|R_{G2} = \frac{R_{C2}\|R_{i3}}{|A_{v2}|} - r_{e2} = 552\,\Omega \;\Rightarrow\; R_{G2} = \frac{1}{\frac{1}{552\,\Omega} - \frac{1}{1\,\text{k}\Omega}} = \mathbf{1.23\,k\Omega}$$

Its input resistance: $R_{i2} = (\beta+1)(r_{e2} + R_{E2}\|R_{G2}) = 58.3\,\text{k}\Omega$.

Step 4 — Input stage (Q_1)

Gain left for stage 1: $|A_{v1}| = \frac{|A_v|}{|A_{v2}|A_{v3}} = \frac{120}{7.893} = 15.2$. The base of Q_2 fixes the collector of Q_1: $V_{RC1} = 1.7\,\text{V}$. Its collector swing is only $\hat{V}_o/(\text{gain after it}) = 0.228\,\text{V}$, so $V_{RE1} \le V_{CC} - V_{RC1} - V_{CE,sat} - 0.228 = 6.872\,\text{V}$; with $V_{RE1} \ge 1\,\text{V}$ use $V_{RE1} = 1\,\text{V}$.

Divider: $V_{RB2} = V_{B1} = 1.7\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 7.3\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \ge R_{i,spec} \;\Rightarrow\; I_{C1} \le \frac{\beta/R_{i,spec}}{\frac{10}{7.3} + \frac{10}{1.7} + \frac{15.2}{1.7}} = 2.06\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 1.8\,\text{mA}$.

Check $I_{C1} \ge N\frac{I_{C2}}{\beta} = 100\,\mu\text{A}$ ✓.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{1.7}{1.8\,\text{mA}} = \mathbf{944\,\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{1.8\,\text{mA}} = \mathbf{556\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 13.9\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 15.2$ with $R_{C1}\|R_{i2} = 929\,\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 47.2\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{47.2\,\Omega} - \frac{1}{556\,\Omega}} = \mathbf{51.6\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{7.3}{180\,\mu\text{A}} = \mathbf{40.6\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{1.7}{180\,\mu\text{A}} = \mathbf{9.44\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 40.6\,\text{k}\Omega\|9.44\,\text{k}\Omega\|6.17\,\text{k}\Omega = 3.419\,\text{k}\Omega \ge 3\,\text{k}\Omega$ ✓.

Step 5 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\|V_{CE}\|$
Q_1	1.8 mA	1.57 mA	6.646 V
Q_2	1 mA	766 μA	4.003 V
Q_3	9.46 mA	6.23 mA	5.477 V

The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 111.8$ instead of 120 (the hand design ignored α and the base currents). As the

slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 111.8/120 (exact $r_{e1} = 15.9\,\Omega$):

$$R_{E1}\|R_{G1} = (15.9\,\Omega + 47.2\,\Omega)\frac{111.8}{120} - 15.9\,\Omega = 42.9\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{46.5\,\Omega}$$

Re-analysis: $|A_v| = 120$ ✓.

The follower would clip at $\hat{V}_o = I_{E3}(R_H\|R_L) = 6.23\,\text{mA} \times 195\,\Omega = 1.217\,\text{V} < 1.5\,\text{V}$: its exact current is below the design value because the base currents shift the DC voltages along the direct-coupled chain. Adjust: lower R_H to the next E12 value, $R_H = \mathbf{470\,\Omega}$ (more follower current), and analyse again. The exact gain is $|A_v| = 118.3$ instead of 120 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 118.3/120 (exact $r_{e1} = 15.9\,\Omega$):

$$R_{E1}\|R_{G1} = (15.9\,\Omega + 42.9\,\Omega)\frac{118.3}{120} - 15.9\,\Omega = 42.1\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{45.5\,\Omega}$$

Re-analysis: $|A_v| = 120$ ✓.

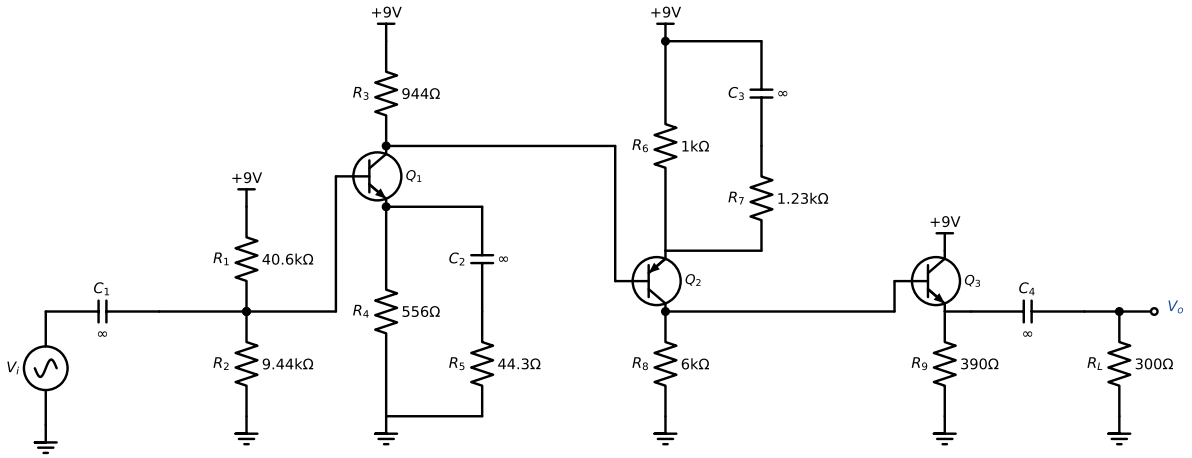
The follower would clip at $\hat{V}_o = I_{E3}(R_H\|R_L) = 7.29\,\text{mA} \times 183\,\Omega = 1.424\,\text{V} < 1.5\,\text{V}$: its exact current is below the design value because the base currents shift the DC voltages along the direct-coupled chain. Adjust: lower R_H to the next E12 value, $R_H = \mathbf{390\,\Omega}$ (more follower current), and analyse again. The exact gain is $|A_v| = 117.9$ instead of 120 (the hand design ignored α and the base currents). As the slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 117.9/120 (exact $r_{e1} = 15.9\,\Omega$):

$$R_{E1}\|R_{G1} = (15.9\,\Omega + 42.1\,\Omega)\frac{117.9}{120} - 15.9\,\Omega = 41.1\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{44.3\,\Omega}$$

Re-analysis: $|A_v| = 120$ ✓.

- Input resistance (exact): $R_i = 3.29\,\text{k}\Omega \geq 3\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E3}(R_H\|R_L) = 1.677\,\text{V}$; Q_2 clips (saturation or cutoff) at $\hat{V}_o = 3.485\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 11.23\,\text{V}$. The smallest, 1.677 V, is $\geq 1.5\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e+R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T|A_v|}{4V_{RC}^2}$): THD $\approx 0.8 + 0.57\,\% = 1.4\,\% \leq 5\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V}$, $V_{RE2} = 1\,\text{V} \geq 1\,\text{V}$ ✓

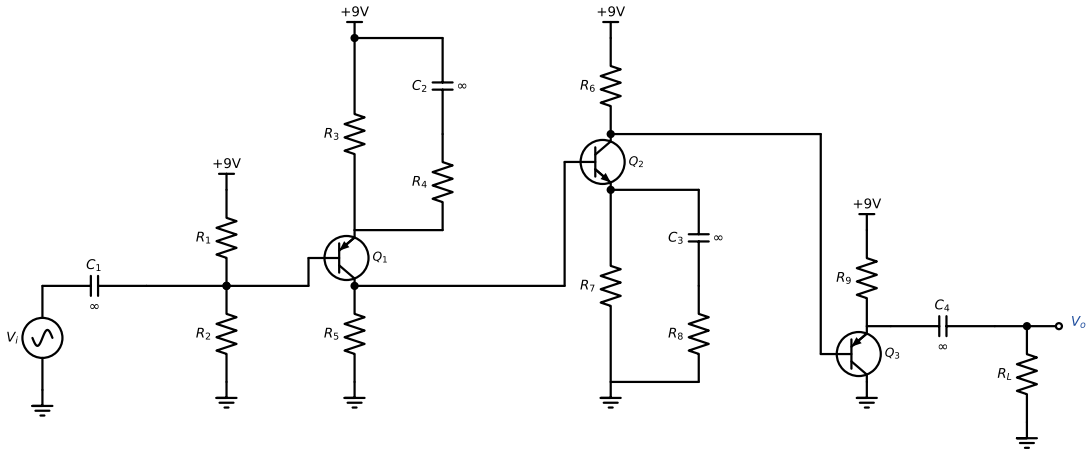
resistor	value
R_1 (R_{B1})	40.6 kΩ
R_2 (R_{B2})	9.44 kΩ
R_3 (R_{C1})	944 Ω
R_4 (R_{E1})	556 Ω
R_5 (R_{G1})	44.3 Ω
R_6 (R_{E2})	1 kΩ
R_7 (R_{G2})	1.23 kΩ
R_8 (R_{C2})	6 kΩ
R_9 (R_H)	390 Ω



Answer

$R_1 = 40.6\,\text{k}\Omega$, $R_2 = 9.44\,\text{k}\Omega$, $R_3 = 944\,\Omega$, $R_4 = 556\,\Omega$, $R_5 = 44.3\,\Omega$, $R_6 = 1\,\text{k}\Omega$, $R_7 = 1.23\,\text{k}\Omega$, $R_8 = 6\,\text{k}\Omega$, $R_9 = 390\,\Omega$

Solution 4E.17 **Hard**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{E1}$, $R_4 = R_{G1}$, $R_5 = R_{C1}$, $R_6 = R_{C2}$, $R_7 = R_{E2}$, $R_8 = R_{G2}$, $R_9 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE2} \geq 1\,\text{V}$; use $V_{RE2} = 1\,\text{V}$.

Collector swing of Q_2: about $\hat{V}_o$ (the follower has a gain close to 1). Largest V_{RC2} without clipping:

$$V_{RC2} = V_{CC} - \hat{V}_o - V_{RE2} - V_{CE,sat} = 9 - 2.5 - 1 - 0.2 = 5.3\,\text{V}$$

Follower Q_3: its emitter sits at 0.7 V from the collector of Q_2, so $V_{RH} = V_{RC2} - 0.7 = 4.6\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC2} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC2} - 0.7 - \hat{V}_o}{\hat{V}_o} = 200\,\Omega \cdot \frac{4.6 - 2.5}{2.5} = 168\,\Omega$$

Take the E12 value just below: $R_H = \mathbf{150\,\Omega}$. Then $I_{C3} \approx \frac{V_{RH}}{R_H} = \frac{4.6}{150\,\Omega} = 30.7\,\text{mA}$, $r_{e3} = \frac{V_T}{I_{C3}} = 815\,\text{m}\Omega$, and with $R_H \| R_L = 85.7\,\Omega$:

$$A_{v3} = \frac{R_H \| R_L}{r_{e3} + R_H \| R_L} = 0.9906, \qquad R_{i3} = (\beta+1)(r_{e3} + R_H \| R_L) = 8.74\,\text{k}\Omega$$

Step 3 — Stage 2 (common emitter, Q_2)

β-insensitive: $I_{C2} \geq N \frac{I_{C3}}{\beta} = 20 \cdot \frac{30.7\,\text{mA}}{100} = 6.13\,\text{mA}$; take the E12 value just above, $I_{C2} = 6.8\,\text{mA}$.

$$R_{C2} = \frac{V_{RC2}}{I_{C2}} = \frac{5.3}{6.8\,\text{mA}} = \mathbf{779\,\Omega}, \quad R_{E2} = \frac{V_{RE2}}{I_{C2}} = \frac{1}{6.8\,\text{mA}} = \mathbf{147\,\Omega}, \quad r_{e2} = \frac{V_T}{I_{C2}} = 3.68\,\Omega$$

Gain $|A_{v2}| = \frac{R_{C2}\|R_{i3}}{r_{e2}+R_{E2}\|R_{G2}} = 8$ with $R_{C2}\|R_{i3} = 716\,\Omega$:

$$R_{E2}\|R_{G2} = \frac{R_{C2}\|R_{i3}}{|A_{v2}|} - r_{e2} = 85.8\,\Omega \;\Rightarrow\; R_{G2} = \frac{1}{\frac{1}{85.8\,\Omega} - \frac{1}{147\,\Omega}} = \mathbf{206\,\Omega}$$

Its input resistance: $R_{i2} = (\beta+1)(r_{e2} + R_{E2}\|R_{G2}) = 9.03\,\mathrm{k}\Omega$.

Step 4 — Input stage (Q_1)

Gain left for stage 1: $|A_{v1}| = \frac{|A_v|}{|A_{v2}|A_{v3}} = \frac{150}{7.925} = 18.93$. The base of Q_2 fixes the collector of Q_1: $V_{RC1} = 1.7\,\mathrm{V}$. Its collector swing is only $\hat{V}_o/(\text{gain after it}) = 0.3155\,\mathrm{V}$, so $V_{RE1} \le V_{CC} - V_{RC1} - V_{CE,sat} - 0.315 = 6.785\,\mathrm{V}$; with $V_{RE1} \ge 1\,\mathrm{V}$ use $V_{RE1} = 1.5\,\mathrm{V}$.

Divider: $V_{RB2} = V_{B1} = 6.8\,\mathrm{V}$, $V_{RB1} = V_{CC} - V_{B1} = 2.2\,\mathrm{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 20$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \ge R_{i,spec} \;\Rightarrow\; I_{C1} \le \frac{\beta/R_{i,spec}}{\frac{20}{2.2} + \frac{20}{6.8} + \frac{18.93}{1.7}} = 2.16\,\mathrm{mA}$$

Take the E12 value just below: $I_{C1} = 1.8\,\mathrm{mA}$.

Check $I_{C1} \ge N\frac{I_{C2}}{\beta} = 1.36\,\mathrm{mA}$ ✓.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{1.7}{1.8\,\mathrm{mA}} = \mathbf{944\,\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1.5}{1.8\,\mathrm{mA}} = \mathbf{833\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 13.9\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 18.93$ with $R_{C1}\|R_{i2} = 855\,\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 31.3\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{31.3\,\Omega} - \frac{1}{833\,\Omega}} = \mathbf{32.5\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{2.2}{360\,\mu\mathrm{A}} = \mathbf{6.11\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{6.8}{360\,\mu\mathrm{A}} = \mathbf{18.9\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 6.11\,\mathrm{k}\Omega\|18.9\,\mathrm{k}\Omega\|4.56\,\mathrm{k}\Omega = 2.295\,\mathrm{k}\Omega \ge 2\,\mathrm{k}\Omega$ ✓.

Step 5 — Check by analysis

Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\|V_{CE}\|$
Q_1	1.8 mA	1.69 mA	6.036 V
Q_2	6.8 mA	5.67 mA	3.921 V
Q_3	30.7 mA	23.3 mA	5.463 V

The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 142.7$ instead of 150 (the hand design ignored α and the base currents). As the

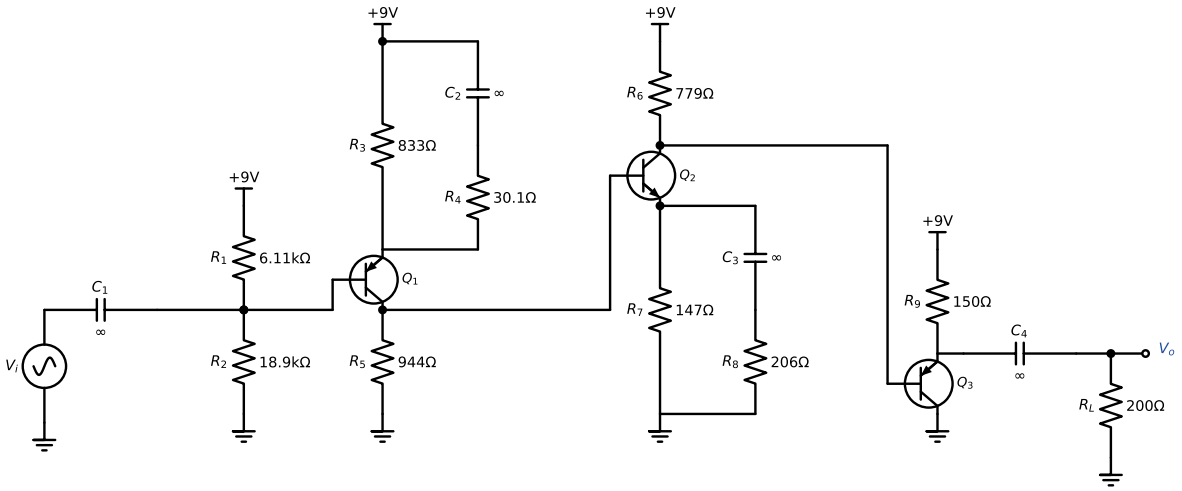
slides suggest, trim R_{G1}: the gain is proportional to $1/(r_{e1} + R_{E1}\|R_{G1})$, so scale that sum by 142.7/150 (exact $r_{e1} = 14.8\,\Omega$):

$$R_{E1}\|R_{G1} = (14.8\,\Omega + 31.3\,\Omega)\frac{142.7}{150} - 14.8\,\Omega = 29\,\Omega \;\Rightarrow\; R_{G1} = \mathbf{30.1\,\Omega}$$

Re-analysis: $|A_v| = 150$ ✓.

- Input resistance (exact): $R_i = 2.26\,\text{k}\Omega \geq 2\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E3}(R_H\|R_L) = 2.001\,\text{V}$; Q_2 clips (saturation or cutoff) at $\hat{V}_o = 3.675\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 11.22\,\text{V}$. The smallest, 2.001 V, is $\geq 2\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e + R_E\|R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T |A_v|}{4V_{RC}^2}$): THD $\approx 1.4 + 0.59\,\% = 2\,\% \leq 5\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1.5\,\text{V}$, $V_{RE2} = 1\,\text{V} \geq 1\,\text{V}$ ✓

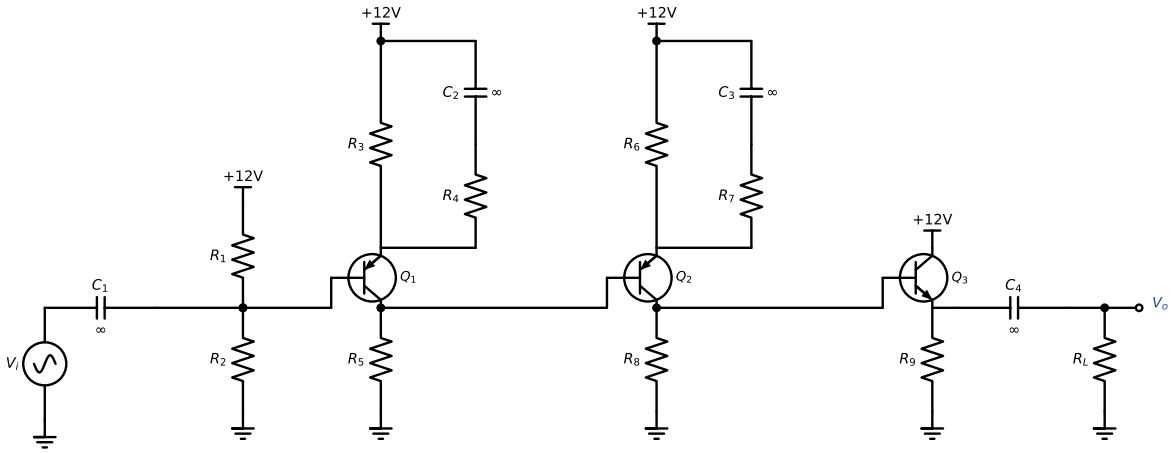
resistor	value
R_1 (R_{B1})	$6.11\,\text{k}\Omega$
R_2 (R_{B2})	$18.9\,\text{k}\Omega$
R_3 (R_{E1})	$833\,\Omega$
R_4 (R_{G1})	$30.1\,\Omega$
R_5 (R_{C1})	$944\,\Omega$
R_6 (R_{C2})	$779\,\Omega$
R_7 (R_{E2})	$147\,\Omega$
R_8 (R_{G2})	$206\,\Omega$
R_9 (R_H)	$150\,\Omega$



Answer

$R_1 = 6.11\,\mathrm{k\Omega}$, $R_2 = 18.9\,\mathrm{k\Omega}$, $R_3 = 833\,\Omega$, $R_4 = 30.1\,\Omega$, $R_5 = 944\,\Omega$, $R_6 = 779\,\Omega$, $R_7 = 147\,\Omega$, $R_8 = 206\,\Omega$, $R_9 = 150\,\Omega$

Solution 4E.18 **Hard**



Step 1 — Plan

Name the resistors by their job: $R_1 = R_{B1}$, $R_2 = R_{B2}$, $R_3 = R_{E1}$, $R_4 = R_{G1}$, $R_5 = R_{C1}$, $R_6 = R_{E2}$, $R_7 = R_{G2}$, $R_8 = R_{C2}$, $R_9 = R_H$. Every stage is biased like the one-stage design of the slides: $V_{RE} \geq 1\,\text{V}$ (V_{BE}-insensitive), the divider and every collector current much larger than the base current it feeds (β-insensitive, factor N), and V_{RC} as large as the swing allows. Small-signal tools: $r_e = V_T/I_C$, CE gain $|A_v| = \frac{Z_C}{r_e + R_E \| R_G}$, follower gain $\frac{Z_E}{r_e + Z_E}$, base resistance $(\beta+1)(r_e + Z_E)$ (the ∞ capacitors put R_G in parallel with R_E for AC only). Design from the output back to the input, because each stage loads the one before it.

Step 2 — Output stage and follower

V_{BE}-insensitive: $V_{RE2} \geq 1\,\text{V}$; use $V_{RE2} = 1\,\text{V}$.

Collector swing of Q_2: about $\hat{V}_o$ (the follower has a gain close to 1). Largest V_{RC2} without clipping:

$$V_{RC2} = V_{CC} - \hat{V}_o - V_{RE2} - V_{CE,sat} = 12 - 3.5 - 1 - 0.2 = 7.3\,\text{V}$$

Follower Q_3: its emitter sits at 0.7 V from the collector of Q_2, so $V_{RH} = V_{RC2} - 0.7 = 6.6\,\text{V}$. To avoid current clipping the DC current in R_H must stay larger than the peak current pushed into R_L:

$$\frac{V_{RC2} - 0.7 - \hat{V}_o}{R_H} \geq \frac{\hat{V}_o}{R_L} \Rightarrow R_H \leq R_L \frac{V_{RC2} - 0.7 - \hat{V}_o}{\hat{V}_o} = 1\,\text{k}\Omega \cdot \frac{6.6 - 3.5}{3.5} = 886\,\Omega$$

Take the E12 value just below: $R_H = \mathbf{820\,\Omega}$. Then $I_{C3} \approx \frac{V_{RH}}{R_H} = \frac{6.6}{820\,\Omega} = 8.05\,\text{mA}$, $r_{e3} = \frac{V_T}{I_{C3}} = 3.11\,\Omega$, and with $R_H \| R_L = 451\,\Omega$:

$$A_{v3} = \frac{R_H \| R_L}{r_{e3} + R_H \| R_L} = 0.9932, \qquad R_{i3} = (\beta+1)(r_{e3} + R_H \| R_L) = 45.8\,\text{k}\Omega$$

Step 3 — Stage 2 (common emitter, Q_2)

β-insensitive: $I_{C2} \geq N \frac{I_{C3}}{\beta} = 10 \cdot \frac{8.05\,\text{mA}}{100} = 805\,\mu\text{A}$; take the E12 value just above, $I_{C2} = 820\,\mu\text{A}$.

$$R_{C2} = \frac{V_{RC2}}{I_{C2}} = \frac{7.3}{820\,\mu\text{A}} = \mathbf{8.9\,\text{k}\Omega}, \quad R_{E2} = \frac{V_{RE2}}{I_{C2}} = \frac{1}{820\,\mu\text{A}} = \mathbf{1.22\,\text{k}\Omega}, \quad r_{e2} = \frac{V_T}{I_{C2}} = 30.5\,\Omega$$

Gain $|A_{v2}| = \frac{R_{C2}\|R_{i3}}{r_{e2}+R_{E2}\|R_{G2}} = 12$ with $R_{C2}\|R_{i3} = 7.45\,\text{k}\Omega$:

$$R_{E2}\|R_{G2} = \frac{R_{C2}\|R_{i3}}{|A_{v2}|} - r_{e2} = 591\,\Omega \;\Rightarrow\; R_{G2} = \frac{1}{\frac{1}{591\,\Omega} - \frac{1}{1.22\,\text{k}\Omega}} = \mathbf{1.15\,k\Omega}$$

Its input resistance: $R_{i2} = (\beta+1)(r_{e2} + R_{E2}\|R_{G2}) = 62.7\,\text{k}\Omega$.

Step 4 — Input stage (Q_1)

Gain left for stage 1: $|A_{v1}| = \frac{|A_v|}{|A_{v2}|A_{v3}} = \frac{300}{11.92} = 25.17$. The base of Q_2 fixes the collector of Q_1: $V_{RC1} = 10.3\,\text{V}$. Its collector swing is only $\hat{V}_o/(\text{gain after it}) = 0.2937\,\text{V}$, so $V_{RE1} \le V_{CC} - V_{RC1} - V_{CE,sat} - 0.294 = 1.206\,\text{V}$; with $V_{RE1} \ge 1\,\text{V}$ use $V_{RE1} = 1\,\text{V}$.

Divider: $V_{RB2} = V_{B1} = 10.3\,\text{V}$, $V_{RB1} = V_{CC} - V_{B1} = 1.7\,\text{V}$ ($|V_{RE1}| + 0.7$ on the emitter side). β-insensitive: the divider current is N times the base current, $I_{RB} = N\frac{I_{C1}}{\beta}$, $N = 10$. Input resistance (all three in parallel, with $(\beta+1)(r_e + R_E\|R_G) \approx \beta\frac{R_{C1}}{|A_{v1}|} = \beta\frac{V_{RC1}}{I_{C1}|A_{v1}|}$):

$$R_i \approx \frac{\beta/I_{C1}}{\frac{N}{V_{RB1}} + \frac{N}{V_{RB2}} + \frac{|A_{v1}|}{V_{RC1}}} \ge R_{i,spec} \;\Rightarrow\; I_{C1} \le \frac{\beta/R_{i,spec}}{\frac{10}{1.7} + \frac{10}{10.3} + \frac{25.17}{10.3}} = 5.38\,\text{mA}$$

Take the E12 value just below: $I_{C1} = 4.7\,\text{mA}$.

Check $I_{C1} \ge N\frac{I_{C2}}{\beta} = 82\,\mu\text{A}$ ✓.

$$R_{C1} = \frac{V_{RC1}}{I_{C1}} = \frac{10.3}{4.7\,\text{mA}} = \mathbf{2.19\,k\Omega}, \quad R_{E1} = \frac{V_{RE1}}{I_{C1}} = \frac{1}{4.7\,\text{mA}} = \mathbf{213\,\Omega}, \quad r_{e1} = \frac{V_T}{I_{C1}} = 5.32\,\Omega$$

Gain $|A_{v1}| = \frac{R_{C1}\|R_{i2}}{r_{e1}+R_{E1}\|R_{G1}} = 25.17$ with $R_{C1}\|R_{i2} = 2.12\,\text{k}\Omega$:

$$R_{E1}\|R_{G1} = \frac{R_{C1}\|R_{i2}}{|A_{v1}|} - r_{e1} = 78.8\,\Omega \;\Rightarrow\; R_{G1} = \frac{1}{\frac{1}{78.8\,\Omega} - \frac{1}{213\,\Omega}} = \mathbf{125\,\Omega}$$

$$R_{B1} = \frac{V_{RB1}}{NI_{C1}/\beta} = \frac{1.7}{470\,\mu\text{A}} = \mathbf{3.62\,k\Omega}, \qquad R_{B2} = \frac{V_{RB2}}{NI_{C1}/\beta} = \frac{10.3}{470\,\mu\text{A}} = \mathbf{21.9\,k\Omega}$$

Check: $R_i = R_{B1}\|R_{B2}\|(\beta+1)(r_{e1} + R_{E1}\|R_{G1}) = 3.62\,\text{k}\Omega\|21.9\,\text{k}\Omega\|8.5\,\text{k}\Omega = 2.274\,\text{k}\Omega \ge 2\,\text{k}\Omega$ ✓.

Step 5 — Check by analysis

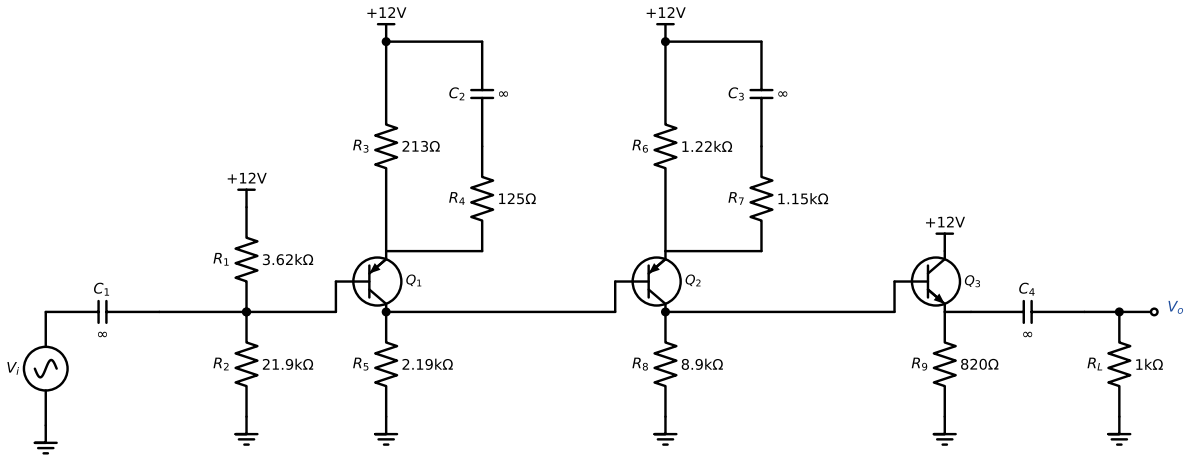
Solve the finished circuit exactly ($\beta = 100$, base currents included) for the DC point, then the small-signal gain with the exact r_e values:

	design I_C	exact I_C	exact $\|V_{CE}\|$
Q_1	4.7 mA	4.07 mA	2.174 V
Q_2	820 µA	1.91 mA	5.736 V
Q_3	8.05 mA	17.7 mA	2.689 V

The exact currents are a little off the design values: N only makes each base current N times smaller than the current it disturbs, so the base voltages sag slightly, and along direct-coupled stages each shift is passed on (and amplified) to the next stage. The slides handle this the same way: simulate, then adjust. The exact gain is $|A_v| = 300.2$, within 0.5 % of the specification ✓.

- Input resistance (exact): $R_i = 2.28\,\text{k}\Omega \geq 2\,\text{k}\Omega$ ✓
- Output swing: follower current clipping at $\hat{V}_o = I_{E3}(R_H \| R_L) = 7.991\,\text{V}$; Q_2 clips (saturation or cutoff) at $\hat{V}_o = 5.519\,\text{V}$; Q_1 clips (saturation or cutoff) at $\hat{V}_o = 24.04\,\text{V}$. The smallest, $5.519\,\text{V}$, is $\geq 3\,\text{V}$ ✓
- Distortion: each CE stage adds $HD_2 \approx \frac{\hat{v}_b}{4V_T}\left(\frac{r_e}{r_e + R_E \| R_G}\right)^2$ ($\hat{v}_b$ = its input amplitude at the specified output; for one stage this is the slides' $\frac{\hat{V}_o V_T |A_v|}{4V_{RC}^2}$): THD $\approx 0.052 + 0.12\,\% = 0.17\,\% \leq 5\,\%$ ✓
- V_{BE}-insensitive: $V_{RE1} = 1\,\text{V}$, $V_{RE2} = 1\,\text{V} \geq 1\,\text{V}$ ✓

resistor	value
R_1 (R_{B1})	$3.62\,\text{k}\Omega$
R_2 (R_{B2})	$21.9\,\text{k}\Omega$
R_3 (R_{E1})	$213\,\Omega$
R_4 (R_{G1})	$125\,\Omega$
R_5 (R_{C1})	$2.19\,\text{k}\Omega$
R_6 (R_{E2})	$1.22\,\text{k}\Omega$
R_7 (R_{G2})	$1.15\,\text{k}\Omega$
R_8 (R_{C2})	$8.9\,\text{k}\Omega$
R_9 (R_H)	$820\,\Omega$



Answer

$R_1 = 3.62\,\text{k}\Omega$, $R_2 = 21.9\,\text{k}\Omega$, $R_3 = 213\,\Omega$, $R_4 = 125\,\Omega$, $R_5 = 2.19\,\text{k}\Omega$, $R_6 = 1.22\,\text{k}\Omega$, $R_7 = 1.15\,\text{k}\Omega$, $R_8 = 8.9\,\text{k}\Omega$, $R_9 = 820\,\Omega$